

# The `ammistability` Package: A Brief Introduction

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## Overview

The package `ammistability` (Ajay et al., 2019a) is a collection of functions for the computation of various stability parameters from the results of Additive Main Effects and Multiplicative Interaction (AMMI) analysis computed by the `AMMI` function of `agricolae` package.

The goal of this vignette is to introduce the users to these functions and give a primer in computation of various stability parameters/indices from a fitted AMMI model. This document assumes a basic knowledge of R programming language.



## Installation

The package can be installed from CRAN as follows:

```
# Install from CRAN
install.packages('ammistability', dependencies=TRUE)
```

The development version can be installed from github as follows:

```
# Install development version from Github
devtools::install_github("ajaygpb/ammistability")
```

Then the package can be loaded using the function

```
library(ammistability)
```

```
-----
Welcome to ammistability version 0.1.4.9000
```

```
# To know how to use this package type:
  browseVignettes(package = 'ammistability')
  for the package vignette.

# To know whats new in this version type:
  news(package='ammistability')
  for the NEWS file.

# To cite the methods in the package type:
  citation(package='ammistability')

# To suppress this message use:
  suppressPackageStartupMessages(library(ammistability))
-----
```

## Version History

The current version of the package is 0.1.4. The previous versions are as follows.

**Table 1.** Version history of ammistability R package.

| Version | Date       |
|---------|------------|
| 0.1.0   | 2018-08-13 |
| 0.1.1   | 2018-12-07 |
| 0.1.2   | 2021-02-23 |
| 0.1.3   | 2022-07-18 |

To know detailed history of changes use `news(package='ammistability')`.

## AMMI model

The difference in response of genotypes to different environmental conditions is known as Genotype-Environment Interaction (GEI). Understanding the nature and structure of this interaction is critical for plant breeders to select for genotypes with wide or specific adaptability. One of the most popular

techniques to achieve this is by fitting the Additive Main Effects and Multiplicative Interaction (AMMI) model to the results of multi environment trials (Gauch, 1988, 1992).

The AMMI equation is described as follows.

$$Y_{ij} = \mu + \alpha_i + \beta_j + \sum_{n=1}^N \lambda_n \gamma_{in} \delta_{jn} + \rho_{ij}$$

Where,  $Y_{ij}$  is the yield of the  $i$ th genotype in the  $j$ th environment,  $\mu$  is the grand mean,  $\alpha_i$  is the genotype deviation from the grand mean,  $\beta_j$  is the environment deviation,  $N$  is the total number of interaction principal components (IPCs),  $\lambda_n$  is the singular value for  $n$ th IPC and correspondingly  $\lambda_n^2$  is its eigen value,  $\gamma_{in}$  is the eigenvector value for  $i$ th genotype,  $\delta_{jn}$  is the eigenvector value for the  $j$ th environment and  $\rho_{ij}$  is the residual.

### AMMI stability parameters

Although the AMMI model can aid in determining genotypes with wide or specific adaptability, it fails to rank genotypes according to their stability. Several measures have been developed over the years to indicate the stability of genotypes from the results of AMMI analysis (Table 1.).

The details about AMMI stability parameters/indices implemented in **ammistability** are described in Table 1.

**Table 1 :** AMMI stability parameters/indices implemented in `ammistability`.

| AMMI stability parameter                                                                             | function                                                      | Details                                                                                                                                   | Reference                                                                    |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Sum across environments of GEI modelled by AMMI ( $AMGE$ )                                           | <code>AMGE.AMMI</code>                                        | $AMGE = \sum_{j=1}^E \sum_{n=1}^{N'} \lambda_n \gamma_{in} \delta_{jn}$                                                                   | Sneller et al. (1997)                                                        |
| AMMI Stability Index ( $ASI$ )                                                                       | <code>ASI.AMMI</code> and <code>MASI.AMMI</code>              | $ASI = \sqrt{[PC_1^2 \times \theta_1^2] + [PC_2^2 \times \theta_2^2]}$                                                                    | Jambhulkar et al. (2014); Jambhulkar et al. (2015); Jambhulkar et al. (2017) |
| AMMI Based Stability Parameter ( $ASTAB$ )                                                           | <code>ASTAB.AMMI</code>                                       | $ASTAB = \sum_{n=1}^{N'} \lambda_n \gamma_{in}^2$                                                                                         | Rao and Prabhakaran (2005)                                                   |
| AMMI stability value ( $ASV$ ) *                                                                     | <code>agricolae::index.AMMI</code> and <code>MASV.AMMI</code> | Distance from the coordinate point to the origin in a two dimensional scattergram generated by plotting of IPC1 score against IPC2 score. | Purchase (1997); Purchase et al. (1999); Purchase et al. (2000)              |
| $ASV = \sqrt{\left(\frac{SSIPC_1}{SSIPC_2} \times PC_1\right)^2 + (PC_2)^2}$                         |                                                               |                                                                                                                                           |                                                                              |
| $AV_{(AMGE)}$                                                                                        | <code>AVAMGE.AMMI</code>                                      | $AV_{(AMGE)} = \sum_{j=1}^E \sum_{n=1}^{N'}  \lambda_n \gamma_{in} \delta_{jn} $                                                          | Zali et al. (2012)                                                           |
| Annicchiarico's D parameter ( $D_a$ )                                                                | <code>DA.AMMI</code>                                          | The unsquared Euclidean distance from the origin of significant IPC axes in the AMMI model.                                               | Annicchiarico (1997)                                                         |
| $D_a = \sqrt{\sum_{n=1}^{N'} (\lambda_n \gamma_{in})^2}$                                             |                                                               |                                                                                                                                           |                                                                              |
| Zhang's D parameter or AMMI statistic coefficient or AMMI distance or AMMI stability index ( $D_z$ ) | <code>DZ.AMMI</code>                                          | The distance of IPC point from origin in space.                                                                                           | Zhang et al. (1998)                                                          |
| $D_z = \sqrt{\sum_{n=1}^{N'} \gamma_{in}^2}$                                                         |                                                               |                                                                                                                                           |                                                                              |
| Averages of the squared eigenvector values $EV$                                                      | <code>EV.AMMI</code>                                          | $EV = \sum_{n=1}^{N'} \frac{\gamma_{in}^2}{N'}$                                                                                           | Zobel (1994)                                                                 |
| Stability measure based on fitted AMMI model $FA$                                                    | <code>FA.AMMI</code>                                          | $FA = \sum_{n=1}^{N'} \lambda_n^2 \gamma_{in}^2$                                                                                          | Raju (2002); Zali et al. (2012)                                              |

| AMMI stability parameter                 | function               | Details                                                                                                                                                                                                                                                                                                                                        | Reference                                      |
|------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| $FP$                                     | <code>FA.AMMI</code>   | Equivalent to $FA$ , when only the first IPC axis is considered for computation.<br><br>$FP = \lambda_1^2 \gamma_{i1}^2$<br><br>As $\lambda_1^2$ will be same for all the genotypes, the absolute value of $\gamma_{i1}$ alone is sufficient for comparison. So this is also equivalent to the comparison based on biplot with first IPC axis. | Raju (2002); Zali et al. (2012)                |
| $B$                                      | <code>FA.AMMI</code>   | Equivalent to $FA$ , when only the first two IPC axes are considered for computation.<br><br>$B = \sum_{n=1}^2 \lambda_n^2 \gamma_{in}^2$<br><br>Stability comparisons based on this measure will be equivalent to the comparisons based on biplot with first two IPC axes.                                                                    | Raju (2002); Zali et al. (2012)                |
| $W_{(AMMI)}$                             | <code>FA.AMMI</code>   | Equivalent to $FA$ , when all the IPC axes in the AMMI model are considered for computation.<br><br>$W_{(AMMI)} = \sum_{n=1}^N \lambda_n^2 \gamma_{in}^2$                                                                                                                                                                                      | Wricke (1962); Raju (2002); Zali et al. (2012) |
| Modified AMMI Stability Index ( $MASI$ ) | <code>MASI.AMMI</code> | Equivalent to Wricke's ecovalence.<br>$MASI = \sqrt{\sum_{n=1}^{N'} PC_n^2 \times \theta_n^2}$                                                                                                                                                                                                                                                 | Ajay et al. (2018)                             |
| Modified AMMI stability value ( $MASV$ ) | <code>MASV.AMMI</code> | $MASV = \sqrt{\sum_{n=1}^{N'-1} \left( \frac{SSIPC_n}{SSIPC_{n+1}} \times PC_n \right)^2 + (PC_{N'})^2}$                                                                                                                                                                                                                                       | Ajay et al. (2019b); Zali et al. (2012)        |

| AMMI stability parameter                                                             | function         | Details                                                                                             | Reference             |
|--------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------|-----------------------|
| Sums of the absolute value of the IPC scores ( <i>SIPC</i> )                         | <b>SIPC.AMMI</b> | $SIPC = \sum_{n=1}^{N'} \left  \lambda_n^{0.5} \gamma_{in} \right $ $SIPC = \sum_{n=1}^{N'}  PC_n $ | Sneller et al. (1997) |
| Absolute value of the relative contribution of IPCs to the interaction ( <i>Za</i> ) | <b>ZA.AMMI</b>   | $Za = \sum_{i=1}^{N'}  \theta_n \gamma_{in} $                                                       | Zali et al. (2012)    |

Where,  $N$  is the total number of interaction principal components (IPCs);  $N'$  is the number of significant IPCAs (number of IPC that were retained in the AMMI model via F tests);  $\lambda_n$  is the singular value for  $n$ th IPC and correspondingly  $\lambda_n^2$  is its eigen value;  $\gamma_{in}$  is the eigenvector value for  $i$ th genotype;  $\delta_{jn}$  is the eigenvector value for the  $j$ th environment;  $SSIPC_1, SSIPC_2, \dots, SSIPC_n$  are the sum of squares of the 1st, 2th,  $\dots$ , and  $n$ th IPC;  $PC_1, PC_2, \dots, PC_n$  are the scores of 1st, 2th,  $\dots$ , and  $n$ th IPC;  $\theta_n$  is the percentage sum of squares explained by  $n$ th principal component interaction effect; and  $E$  is the number of environments.

## Examples

```
library(agricolae)
data(plrv)

# AMMI model
model <- with(plrv, AMMI(Locality, Genotype, Rep, Yield, console = FALSE))

# ANOVA
model$ANOVA
```

## AMMI model from agricolae::AMMI

## Analysis of Variance Table

Response: Y

|           | Df  | Sum Sq | Mean Sq | F value  | Pr(>F)        |
|-----------|-----|--------|---------|----------|---------------|
| ENV       | 5   | 122284 | 24456.9 | 257.0382 | 9.08e-12 ***  |
| REP(ENV)  | 12  | 1142   | 95.1    | 2.5694   | 0.002889 **   |
| GEN       | 27  | 17533  | 649.4   | 17.5359  | < 2.2e-16 *** |
| ENV:GEN   | 135 | 23762  | 176.0   | 4.7531   | < 2.2e-16 *** |
| Residuals | 324 | 11998  | 37.0    |          |               |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

```
# IPC F test
model$analysis
```

|     | percent | acum  | Df | Sum.Sq     | Mean.Sq   | F.value | Pr.F   |
|-----|---------|-------|----|------------|-----------|---------|--------|
| PC1 | 56.3    | 56.3  | 31 | 13368.5954 | 431.24501 | 11.65   | 0.0000 |
| PC2 | 27.1    | 83.3  | 29 | 6427.5799  | 221.64069 | 5.99    | 0.0000 |
| PC3 | 9.4     | 92.7  | 27 | 2241.9398  | 83.03481  | 2.24    | 0.0005 |
| PC4 | 4.3     | 97.1  | 25 | 1027.5785  | 41.10314  | 1.11    | 0.3286 |
| PC5 | 2.9     | 100.0 | 23 | 696.1012   | 30.26527  | 0.82    | 0.7059 |

```
# Mean yield and IPC scores
model$biplot
```

|        | type | Yield    | PC1         | PC2          | PC3         | PC4         | PC5         |
|--------|------|----------|-------------|--------------|-------------|-------------|-------------|
| 102.18 | GEN  | 26.31947 | -1.50828851 | 1.258765244  | -0.19220309 | 0.48738861  | -0.04364115 |
| 104.22 | GEN  | 31.28887 | 0.32517729  | -1.297024517 | -0.63695749 | -0.44159957 | 0.95312506  |
| 121.31 | GEN  | 30.10174 | 0.95604605  | 1.143461054  | -1.28777348 | 2.22246913  | -1.30661916 |
| 141.28 | GEN  | 39.75624 | 2.11153737  | 0.817810467  | 1.45527701  | 0.25257620  | -0.25996142 |
| 157.26 | GEN  | 36.95181 | 1.05139017  | 2.461179974  | -1.97208942 | -1.96538800 | -0.59719268 |
| 163.9  | GEN  | 21.41747 | -2.12407441 | -0.284381234 | -0.21791137 | -0.50743629 | 0.18563390  |
| 221.19 | GEN  | 22.98480 | -0.84981828 | 0.347983673  | -0.82400783 | -0.11451944 | -0.57504816 |
| 233.11 | GEN  | 28.66655 | 0.07554203  | -1.046497338 | 1.04040485  | 0.22868362  | 0.65754266  |
| 235.6  | GEN  | 38.63477 | 1.20102029  | -2.816581184 | 0.80975361  | 1.02013062  | -0.40273415 |
| 241.2  | GEN  | 26.34039 | -0.79948495 | 0.220768053  | -0.98538801 | 0.30004421  | 0.07555258  |
| 255.7  | GEN  | 30.58975 | -1.49543817 | -1.186549449 | 0.92552519  | -0.32009239 | -0.46344763 |
| 314.12 | GEN  | 28.17335 | 1.39335380  | -0.332786322 | -0.73226877 | 0.05987348  | 0.54406154  |
| 317.6  | GEN  | 35.32583 | 1.05170769  | 0.002555823  | -0.81561907 | 0.58180433  | 0.39627052  |
| 319.20 | GEN  | 38.75767 | 3.08338144  | 1.995946966  | 0.87971668  | -1.11908943 | 0.29657050  |
| 320.16 | GEN  | 26.34808 | -1.55737097 | 0.732314249  | -0.41432567 | 1.32097009  | 2.29506737  |
| 342.15 | GEN  | 26.01336 | -1.35880873 | -0.741980068 | 0.87480105  | -1.12013125 | -0.10776433 |

|         |     |          |             |              |             |             |             |
|---------|-----|----------|-------------|--------------|-------------|-------------|-------------|
| 346.2   | GEN | 23.84175 | -2.48453928 | -0.397045286 | 1.07091711  | -0.90974484 | -0.12738693 |
| 351.26  | GEN | 36.11581 | 1.22670345  | 1.537183139  | 1.79835728  | -0.03516368 | 0.30191335  |
| 364.21  | GEN | 34.05974 | 0.27328985  | -0.447941156 | 0.03139543  | 0.77920500  | -0.95811256 |
| 402.7   | GEN | 27.47748 | -0.12907269 | -0.080086669 | 0.01934016  | -0.36085862 | -0.28473777 |
| 405.2   | GEN | 28.98663 | -1.90936369 | 0.309047963  | 0.57682642  | 0.51163370  | -0.34397623 |
| 406.12  | GEN | 32.68323 | 0.90781100  | -1.733433781 | -0.24223050 | -0.38596144 | -0.49796296 |
| 427.7   | GEN | 36.19020 | 0.42791957  | -0.723190970 | -0.85381724 | -0.53089914 | 1.00677993  |
| 450.3   | GEN | 36.19602 | 1.38026196  | 1.279525147  | 0.16025163  | 0.61270137  | -0.34325251 |
| 506.2   | GEN | 33.26623 | -0.33054261 | -0.302588536 | -1.58471588 | -0.04659416 | 0.87807441  |
| Canchan | GEN | 27.00126 | 1.47802905  | 0.380553178  | 1.67423900  | 0.07718375  | 0.49381313  |
| Desiree | GEN | 16.15569 | -3.64968796 | 1.720025405  | 0.43761089  | 0.04648011  | -0.86767477 |
| Unica   | GEN | 39.10400 | 1.25331924  | -2.817033826 | -0.99510845 | -0.64366599 | -0.90489253 |
| Ayac    | ENV | 23.70254 | -2.29611851 | 0.966037760  | 1.95959116  | 2.75548057  | 1.67177210  |
| Hyo-02  | ENV | 45.73082 | 3.85283195  | -5.093371615 | 1.16967118  | -0.08985538 | 0.01540152  |
| LM-02   | ENV | 34.64462 | -1.14575146 | -0.881093222 | -4.56547274 | 0.55159099  | 0.52350416  |
| LM-03   | ENV | 53.83493 | 5.34625518  | 4.265275487  | -0.14143931 | -0.11714533 | -0.40285728 |
| SR-02   | ENV | 14.95128 | -2.58678337 | 0.660309540  | 0.89096920  | -3.25055305 | 1.37283488  |
| SR-03   | ENV | 11.15328 | -3.17043379 | 0.082842050  | 0.68668051  | 0.15048221  | -3.18065538 |

```
# G*E matrix (deviations from mean)
```

```
array(model$genXenv, dim(model$genXenv), dimnames(model$genXenv))
```

|         | ENV         |             |             |             |              |             |  |
|---------|-------------|-------------|-------------|-------------|--------------|-------------|--|
| GEN     | Ayac        | Hyo-02      | LM-02       | LM-03       | SR-02        | SR-03       |  |
| 102.18  | 5.5726162   | -12.4918224 | 1.7425251   | -2.7070438  | 2.91734869   | 4.9663762   |  |
| 104.22  | -2.8712076  | 7.1684102   | 3.9336218   | -4.0358373  | 0.47881580   | -4.6738028  |  |
| 121.31  | 0.3255230   | -3.8666836  | 4.3182811   | 10.4366135  | -11.88343843 | 0.6697043   |  |
| 141.28  | -0.9451837  | 5.6454825   | -9.7806639  | 14.6463104  | -4.80337115  | -4.7625741  |  |
| 157.26  | -10.3149711 | -10.6241677 | 4.2336365   | 16.8683612  | 2.71710210   | -2.8799609  |  |
| 163.9   | 3.0874931   | -6.9416721  | 3.4963790   | -12.5533271 | 7.01688164   | 5.8942454   |  |
| 221.19  | -0.6041752  | -6.0090018  | 4.0648518   | -2.6974743  | 1.27671246   | 3.9690870   |  |
| 233.11  | 2.5837535   | 6.8277609   | -3.4440645  | -4.4985717  | 0.19989490   | -1.6687730  |  |
| 235.6   | -1.7541523  | 19.8225025  | -2.2394463  | -5.6643239  | -8.11400542  | -2.0505746  |  |
| 241.2   | 1.0710975   | -5.3831118  | 5.4253097   | -3.2588271  | 0.46433086   | 1.6812008   |  |
| 255.7   | 2.4443155   | 1.3860497   | -1.8857757  | -12.9626594 | 4.31373929   | 6.7043306   |  |
| 314.12  | -3.8812099  | 6.2098482   | 2.3577759   | 5.9071782   | -3.92419060  | -6.6694018  |  |
| 317.6   | -1.7450319  | 3.0388540   | 3.0448064   | 5.5211634   | -4.79271565  | -5.0670763  |  |
| 319.20  | -6.0155949  | 2.8477540   | -9.7697504  | 24.8850017  | -1.82949467  | -10.1179157 |  |
| 320.16  | 10.9481796  | -10.2982108 | 4.9608280   | -6.2233088  | 2.99984918   | -2.3873373  |  |
| 342.15  | 0.8508002   | -0.3338618  | -2.4575390  | -10.3783871 | 7.29753151   | 5.0214562   |  |
| 346.2   | 4.7000495   | -6.2178087  | -2.2612391  | -14.9700672 | 9.90123888   | 8.8478267   |  |
| 351.26  | 2.6002030   | -0.9918665  | -10.8315931 | 12.7429121  | -0.02713985  | -3.4925156  |  |
| 364.21  | -0.4533734  | 3.2864208   | -0.1335527  | -0.1592533  | -4.82292664  | 2.2826853   |  |
| 402.7   | -1.2134573  | -0.0387229  | -0.2179557  | -0.8774011  | 1.08032472   | 1.2672123   |  |
| 405.2   | 6.6477681   | -8.3071271  | -0.6159895  | -8.8927189  | 3.52179705   | 7.6462704   |  |
| 406.12  | -6.1296667  | 12.0703469  | 1.1195092   | -2.2601009  | -3.13776595  | -1.6623226  |  |
| 427.7   | -3.1340922  | 4.3967072   | 4.2792028   | -1.0194744  | 0.76266844   | -5.2850119  |  |
| 450.3   | -0.5047010  | -1.0720791  | -3.2821761  | 12.8806007  | -5.04562407  | -2.9760204  |  |
| 506.2   | -1.2991912  | -1.5682154  | 8.3142802   | -3.1819279  | 0.60021498   | -2.8651608  |  |
| Canchan | 1.2929442   | 5.7152780   | -9.3713622  | 9.0803035   | -1.65332869  | -5.0638348  |  |
| Desiree | 9.5767845   | -22.3280421 | 0.2396387   | -11.8935722 | 9.62433886   | 14.7808522  |  |
| Unica   | -10.8355195 | 18.0569790  | 4.7604622   | -4.7341684  | -5.13878822  | -2.1089651  |  |



```
# With default n (N') and default ssi.method (farshadfar)
AMGE.AMMI(model)
```

```
AMGE.AMMI()
```

|         | AMGE          | SSI  | rAMGE | rY | means    |
|---------|---------------|------|-------|----|----------|
| 102.18  | 6.661338e-15  | 44.0 | 21.0  | 23 | 26.31947 |
| 104.22  | 1.110223e-15  | 30.0 | 17.0  | 13 | 31.28887 |
| 121.31  | 1.199041e-14  | 42.0 | 27.0  | 15 | 30.10174 |
| 141.28  | -1.110223e-14 | 4.0  | 3.0   | 1  | 39.75624 |
| 157.26  | 2.131628e-14  | 33.0 | 28.0  | 5  | 36.95181 |
| 163.9   | 5.329071e-15  | 47.0 | 20.0  | 27 | 21.41747 |
| 221.19  | 9.325873e-15  | 49.5 | 23.5  | 26 | 22.98480 |
| 233.11  | -1.088019e-14 | 21.0 | 4.0   | 17 | 28.66655 |
| 235.6   | -1.687539e-14 | 5.0  | 1.0   | 4  | 38.63477 |
| 241.2   | 9.325873e-15  | 45.5 | 23.5  | 22 | 26.34039 |
| 255.7   | -8.215650e-15 | 20.0 | 6.0   | 14 | 30.58975 |
| 314.12  | 0.000000e+00  | 33.0 | 15.0  | 18 | 28.17335 |
| 317.6   | 4.884981e-15  | 28.0 | 19.0  | 9  | 35.32583 |
| 319.20  | -7.105427e-15 | 11.0 | 8.0   | 3  | 38.75767 |
| 320.16  | 7.993606e-15  | 43.0 | 22.0  | 21 | 26.34808 |
| 342.15  | -7.327472e-15 | 31.0 | 7.0   | 24 | 26.01336 |
| 346.2   | -6.217249e-15 | 34.0 | 9.0   | 25 | 23.84175 |
| 351.26  | -8.659740e-15 | 13.0 | 5.0   | 8  | 36.11581 |
| 364.21  | -1.915135e-15 | 22.0 | 12.0  | 10 | 34.05974 |
| 402.7   | -2.636780e-16 | 33.0 | 14.0  | 19 | 27.47748 |
| 405.2   | 4.440892e-16  | 32.0 | 16.0  | 16 | 28.98663 |
| 406.12  | -5.995204e-15 | 22.0 | 10.0  | 12 | 32.68323 |
| 427.7   | 3.219647e-15  | 25.0 | 18.0  | 7  | 36.19020 |
| 450.3   | -4.440892e-16 | 19.0 | 13.0  | 6  | 36.19602 |
| 506.2   | 1.176836e-14  | 37.0 | 26.0  | 11 | 33.26623 |
| Canchan | -1.421085e-14 | 22.0 | 2.0   | 20 | 27.00126 |
| Desiree | 1.088019e-14  | 53.0 | 25.0  | 28 | 16.15569 |
| Unica   | -2.664535e-15 | 13.0 | 11.0  | 2  | 39.10400 |

```
# With n = 4 and default ssi.method (farshadfar)
AMGE.AMMI(model, n = 4)
```

|        | AMGE          | SSI  | rAMGE | rY | means    |
|--------|---------------|------|-------|----|----------|
| 102.18 | 1.088019e-14  | 46.5 | 23.5  | 23 | 26.31947 |
| 104.22 | -2.220446e-15 | 26.0 | 13.0  | 13 | 31.28887 |
| 121.31 | 2.575717e-14  | 43.0 | 28.0  | 15 | 30.10174 |
| 141.28 | -8.770762e-15 | 11.0 | 10.0  | 1  | 39.75624 |
| 157.26 | 8.437695e-15  | 25.0 | 20.0  | 5  | 36.95181 |
| 163.9  | 2.664535e-15  | 43.0 | 16.0  | 27 | 21.41747 |
| 221.19 | 8.493206e-15  | 47.0 | 21.0  | 26 | 22.98480 |
| 233.11 | -9.214851e-15 | 24.0 | 7.0   | 17 | 28.66655 |
| 235.6  | -1.199041e-14 | 9.0  | 5.0   | 4  | 38.63477 |
| 241.2  | 1.088019e-14  | 45.5 | 23.5  | 22 | 26.34039 |
| 255.7  | -1.088019e-14 | 20.0 | 6.0   | 14 | 30.58975 |
| 314.12 | -4.440892e-16 | 32.0 | 14.0  | 18 | 28.17335 |
| 317.6  | 8.881784e-15  | 31.0 | 22.0  | 9  | 35.32583 |
| 319.20 | -1.554312e-14 | 4.0  | 1.0   | 3  | 38.75767 |
| 320.16 | 1.509903e-14  | 48.0 | 27.0  | 21 | 26.34808 |

```

342.15 -1.376677e-14 27.0 3.0 24 26.01336
346.2 -1.243450e-14 29.0 4.0 25 23.84175
351.26 -8.881784e-15 16.5 8.5 8 36.11581
364.21 2.997602e-15 27.0 17.0 10 34.05974
402.7 -2.726985e-15 31.0 12.0 19 27.47748
405.2 5.939693e-15 35.0 19.0 16 28.98663
406.12 -8.881784e-15 20.5 8.5 12 32.68323
427.7 -1.110223e-16 22.0 15.0 7 36.19020
450.3 3.774758e-15 24.0 18.0 6 36.19602
506.2 1.135203e-14 36.0 25.0 11 33.26623
Canchan -1.521006e-14 22.0 2.0 20 27.00126
Desiree 1.343370e-14 54.0 26.0 28 16.15569
Unica -7.993606e-15 13.0 11.0 2 39.10400

```

```

# With default n (N') and ssi.method = "rao"
AMGE.AMMI(model, ssi.method = "rao")

```

|         | AMGE          | SSI       | rAMGE | rY | means    |
|---------|---------------|-----------|-------|----|----------|
| 102.18  | 6.661338e-15  | 0.8581621 | 21.0  | 23 | 26.31947 |
| 104.22  | 1.110223e-15  | 1.0201923 | 17.0  | 13 | 31.28887 |
| 121.31  | 1.199041e-14  | 0.9814852 | 27.0  | 15 | 30.10174 |
| 141.28  | -1.110223e-14 | 1.2962760 | 3.0   | 1  | 39.75624 |
| 157.26  | 2.131628e-14  | 1.2048361 | 28.0  | 5  | 36.95181 |
| 163.9   | 5.329071e-15  | 0.6983294 | 20.0  | 27 | 21.41747 |
| 221.19  | 9.325873e-15  | 0.7494331 | 23.5  | 26 | 22.98480 |
| 233.11  | -1.088019e-14 | 0.9346902 | 4.0   | 17 | 28.66655 |
| 235.6   | -1.687539e-14 | 1.2597100 | 1.0   | 4  | 38.63477 |
| 241.2   | 9.325873e-15  | 0.8588442 | 23.5  | 22 | 26.34039 |
| 255.7   | -8.215650e-15 | 0.9973970 | 6.0   | 14 | 30.58975 |
| 314.12  | 0.000000e+00  | NaN       | 15.0  | 18 | 28.17335 |
| 317.6   | 4.884981e-15  | 1.1518199 | 19.0  | 9  | 35.32583 |
| 319.20  | -7.105427e-15 | 1.2637171 | 8.0   | 3  | 38.75767 |
| 320.16  | 7.993606e-15  | 0.8590950 | 22.0  | 21 | 26.34808 |
| 342.15  | -7.327472e-15 | 0.8481812 | 7.0   | 24 | 26.01336 |
| 346.2   | -6.217249e-15 | 0.7773746 | 9.0   | 25 | 23.84175 |
| 351.26  | -8.659740e-15 | 1.1775777 | 5.0   | 8  | 36.11581 |
| 364.21  | -1.915135e-15 | 1.1105384 | 12.0  | 10 | 34.05974 |
| 402.7   | -2.636780e-16 | 0.8959198 | 14.0  | 19 | 27.47748 |
| 405.2   | 4.440892e-16  | 0.9451263 | 16.0  | 16 | 28.98663 |
| 406.12  | -5.995204e-15 | 1.0656562 | 10.0  | 12 | 32.68323 |
| 427.7   | 3.219647e-15  | 1.1800032 | 18.0  | 7  | 36.19020 |
| 450.3   | -4.440892e-16 | 1.1801930 | 13.0  | 6  | 36.19602 |
| 506.2   | 1.176836e-14  | 1.0846655 | 26.0  | 11 | 33.26623 |
| Canchan | -1.421085e-14 | 0.8803922 | 2.0   | 20 | 27.00126 |
| Desiree | 1.088019e-14  | 0.5267659 | 25.0  | 28 | 16.15569 |
| Unica   | -2.664535e-15 | 1.2750094 | 11.0  | 2  | 39.10400 |

```

# Changing the ratio of weights for Rao's SSI
AMGE.AMMI(model, ssi.method = "rao", a = 0.43)

```

|        | AMGE          | SSI       | rAMGE | rY | means    |
|--------|---------------|-----------|-------|----|----------|
| 102.18 | 6.661338e-15  | 0.8581621 | 21.0  | 23 | 26.31947 |
| 104.22 | 1.110223e-15  | 1.0201923 | 17.0  | 13 | 31.28887 |
| 121.31 | 1.199041e-14  | 0.9814852 | 27.0  | 15 | 30.10174 |
| 141.28 | -1.110223e-14 | 1.2962760 | 3.0   | 1  | 39.75624 |
| 157.26 | 2.131628e-14  | 1.2048361 | 28.0  | 5  | 36.95181 |

```

163.9    5.329071e-15 0.6983294 20.0 27 21.41747
221.19   9.325873e-15 0.7494331 23.5 26 22.98480
233.11  -1.088019e-14 0.9346902   4.0 17 28.66655
235.6   -1.687539e-14 1.2597100   1.0  4 38.63477
241.2    9.325873e-15 0.8588442 23.5 22 26.34039
255.7   -8.215650e-15 0.9973970   6.0 14 30.58975
314.12   0.000000e+00      NaN  15.0 18 28.17335
317.6    4.884981e-15 1.1518199  19.0  9 35.32583
319.20  -7.105427e-15 1.2637171   8.0  3 38.75767
320.16   7.993606e-15 0.8590950  22.0 21 26.34808
342.15  -7.327472e-15 0.8481812   7.0 24 26.01336
346.2   -6.217249e-15 0.7773746   9.0 25 23.84175
351.26  -8.659740e-15 1.1775777   5.0  8 36.11581
364.21  -1.915135e-15 1.1105384  12.0 10 34.05974
402.7   -2.636780e-16 0.8959198  14.0 19 27.47748
405.2    4.440892e-16 0.9451263  16.0 16 28.98663
406.12  -5.995204e-15 1.0656562  10.0 12 32.68323
427.7    3.219647e-15 1.1800032  18.0  7 36.19020
450.3   -4.440892e-16 1.1801930  13.0  6 36.19602
506.2    1.176836e-14 1.0846655  26.0 11 33.26623
Canchan -1.421085e-14 0.8803922   2.0 20 27.00126
Desiree  1.088019e-14 0.5267659  25.0 28 16.15569
Unica   -2.664535e-15 1.2750094  11.0  2 39.10400

```

```

# With default ssi.method (farshadfar)
ASI.AMMI(model)

```

```
ASI.AMMI()
```

```

          ASI SSI rASI rY    means
102.18  0.91512303 43   20 23 26.31947
104.22  0.39631322 19    6 13 31.28887
121.31  0.62108102 25   10 15 30.10174
141.28  1.20927797 26   25  1 39.75624
157.26  0.89176583 22   17  5 36.95181
163.9   1.19833464 51   24 27 21.41747
221.19  0.48765291 34    8 26 22.98480
233.11  0.28677206 21    4 17 28.66655
235.6   1.01971997 25   21  4 38.63477
241.2   0.45406877 29    7 22 26.34039
255.7   0.90124720 33   19 14 30.58975
314.12  0.78962523 30   12 18 28.17335
317.6   0.59211183 18    9  9 35.32583
319.20  1.81826161 30   27  3 38.75767
320.16  0.89897900 39   18 21 26.34808
342.15  0.79099371 37   13 24 26.01336
346.2   1.40292793 51   26 25 23.84175
351.26  0.80654291 22   14  8 36.11581
364.21  0.19598368 12    2 10 34.05974
402.7   0.07583976 20    1 19 27.47748
405.2   1.07822942 39   23 16 28.98663
406.12  0.69418710 23   11 12 32.68323
427.7   0.31056699 12    5  7 36.19020
450.3   0.85094150 22   16  6 36.19602

```

```
506.2  0.20336120  14   3 11 33.26623
Canchan 0.83849670  35  15 20 27.00126
Desiree 2.10698168  56  28 28 16.15569
Unica   1.03956820  24  22  2 39.10400
```

```
# With ssi.method = "rao"
```

```
ASI.AMMI(model, ssi.method = "rao")
```

|         | ASI        | SSI       | rASI | rY | means    |
|---------|------------|-----------|------|----|----------|
| 102.18  | 0.91512303 | 1.3832387 | 20   | 23 | 26.31947 |
| 104.22  | 0.39631322 | 2.2326416 | 6    | 13 | 31.28887 |
| 121.31  | 0.62108102 | 1.7551519 | 10   | 15 | 30.10174 |
| 141.28  | 1.20927797 | 1.6936286 | 25   | 1  | 39.75624 |
| 157.26  | 0.89176583 | 1.7436656 | 17   | 5  | 36.95181 |
| 163.9   | 1.19833464 | 1.0993106 | 24   | 27 | 21.41747 |
| 221.19  | 0.48765291 | 1.7347850 | 8    | 26 | 22.98480 |
| 233.11  | 0.28677206 | 2.6102708 | 4    | 17 | 28.66655 |
| 235.6   | 1.01971997 | 1.7309273 | 21   | 4  | 38.63477 |
| 241.2   | 0.45406877 | 1.9170753 | 7    | 22 | 26.34039 |
| 255.7   | 0.90124720 | 1.5305578 | 19   | 14 | 30.58975 |
| 314.12  | 0.78962523 | 1.5271379 | 12   | 18 | 28.17335 |
| 317.6   | 0.59211183 | 1.9633384 | 9    | 9  | 35.32583 |
| 319.20  | 1.81826161 | 1.5279859 | 27   | 3  | 38.75767 |
| 320.16  | 0.89897900 | 1.3936010 | 18   | 21 | 26.34808 |
| 342.15  | 0.79099371 | 1.4556573 | 13   | 24 | 26.01336 |
| 346.2   | 1.40292793 | 1.1198795 | 26   | 25 | 23.84175 |
| 351.26  | 0.80654291 | 1.7733422 | 14   | 8  | 36.11581 |
| 364.21  | 0.19598368 | 3.5623227 | 2    | 10 | 34.05974 |
| 402.7   | 0.07583976 | 7.2317748 | 1    | 19 | 27.47748 |
| 405.2   | 1.07822942 | 1.3907733 | 23   | 16 | 28.98663 |
| 406.12  | 0.69418710 | 1.7578467 | 11   | 12 | 32.68323 |
| 427.7   | 0.31056699 | 2.7272047 | 5    | 7  | 36.19020 |
| 450.3   | 0.85094150 | 1.7448731 | 16   | 6  | 36.19602 |
| 506.2   | 0.20336120 | 3.4475042 | 3    | 11 | 33.26623 |
| Canchan | 0.83849670 | 1.4534532 | 15   | 20 | 27.00126 |
| Desiree | 2.10698168 | 0.7548219 | 28   | 28 | 16.15569 |
| Unica   | 1.03956820 | 1.7372299 | 22   | 2  | 39.10400 |

```
# Changing the ratio of weights for Rao's SSI
```

```
ASI.AMMI(model, ssi.method = "rao", a = 0.43)
```

|        | ASI        | SSI       | rASI | rY | means    |
|--------|------------|-----------|------|----|----------|
| 102.18 | 0.91512303 | 1.0839450 | 20   | 23 | 26.31947 |
| 104.22 | 0.39631322 | 1.5415455 | 6    | 13 | 31.28887 |
| 121.31 | 0.62108102 | 1.3141619 | 10   | 15 | 30.10174 |
| 141.28 | 1.20927797 | 1.4671376 | 25   | 1  | 39.75624 |
| 157.26 | 0.89176583 | 1.4365328 | 17   | 5  | 36.95181 |
| 163.9  | 1.19833464 | 0.8707513 | 24   | 27 | 21.41747 |
| 221.19 | 0.48765291 | 1.1731344 | 8    | 26 | 22.98480 |
| 233.11 | 0.28677206 | 1.6551898 | 4    | 17 | 28.66655 |
| 235.6  | 1.01971997 | 1.4623334 | 21   | 4  | 38.63477 |
| 241.2  | 0.45406877 | 1.3138836 | 7    | 22 | 26.34039 |
| 255.7  | 0.90124720 | 1.2266562 | 19   | 14 | 30.58975 |
| 314.12 | 0.78962523 | 1.1802765 | 12   | 18 | 28.17335 |
| 317.6  | 0.59211183 | 1.5007728 | 9    | 9  | 35.32583 |
| 319.20 | 1.81826161 | 1.3773527 | 27   | 3  | 38.75767 |

|         |            |           |    |    |          |
|---------|------------|-----------|----|----|----------|
| 320.16  | 0.89897900 | 1.0889326 | 18 | 21 | 26.34808 |
| 342.15  | 0.79099371 | 1.1093959 | 13 | 24 | 26.01336 |
| 346.2   | 1.40292793 | 0.9246517 | 26 | 25 | 23.84175 |
| 351.26  | 0.80654291 | 1.4337564 | 14 | 8  | 36.11581 |
| 364.21  | 0.19598368 | 2.1648057 | 2  | 10 | 34.05974 |
| 402.7   | 0.07583976 | 3.6203374 | 1  | 19 | 27.47748 |
| 405.2   | 1.07822942 | 1.1367545 | 23 | 16 | 28.98663 |
| 406.12  | 0.69418710 | 1.3632981 | 11 | 12 | 32.68323 |
| 427.7   | 0.31056699 | 1.8452998 | 5  | 7  | 36.19020 |
| 450.3   | 0.85094150 | 1.4230055 | 16 | 6  | 36.19602 |
| 506.2   | 0.20336120 | 2.1006861 | 3  | 11 | 33.26623 |
| Canchan | 0.83849670 | 1.1268084 | 15 | 20 | 27.00126 |
| Desiree | 2.10698168 | 0.6248300 | 28 | 28 | 16.15569 |
| Unica   | 1.03956820 | 1.4737642 | 22 | 2  | 39.10400 |

```
# With default n (N') and default ssi.method (farshadfar)
ASTAB.AMMI(model)
```

```
ASTAB.AMMI()
```

|         | ASTAB       | SSI | rASTAB | rY | means    |
|---------|-------------|-----|--------|----|----------|
| 102.18  | 3.89636621  | 39  | 16     | 23 | 26.31947 |
| 104.22  | 2.19372771  | 21  | 8      | 13 | 31.28887 |
| 121.31  | 3.87988776  | 29  | 14     | 15 | 30.10174 |
| 141.28  | 7.24523520  | 23  | 22     | 1  | 39.75624 |
| 157.26  | 11.05196482 | 31  | 26     | 5  | 36.95181 |
| 163.9   | 4.64005014  | 46  | 19     | 27 | 21.41747 |
| 221.19  | 1.52227265  | 30  | 4      | 26 | 22.98480 |
| 233.11  | 2.18330553  | 24  | 7      | 17 | 28.66655 |
| 235.6   | 10.03128021 | 28  | 24     | 4  | 38.63477 |
| 241.2   | 1.65890425  | 27  | 5      | 22 | 26.34039 |
| 255.7   | 4.50083178  | 32  | 18     | 14 | 30.58975 |
| 314.12  | 2.58839912  | 27  | 9      | 18 | 28.17335 |
| 317.6   | 1.77133006  | 15  | 6      | 9  | 35.32583 |
| 319.20  | 14.26494686 | 30  | 27     | 3  | 38.75767 |
| 320.16  | 3.13335427  | 32  | 11     | 21 | 26.34808 |
| 342.15  | 3.16217247  | 36  | 12     | 24 | 26.01336 |
| 346.2   | 7.47744386  | 48  | 23     | 25 | 23.84175 |
| 351.26  | 7.10182225  | 29  | 21     | 8  | 36.11581 |
| 364.21  | 0.27632429  | 12  | 2      | 10 | 34.05974 |
| 402.7   | 0.02344768  | 20  | 1      | 19 | 27.47748 |
| 405.2   | 4.07390905  | 33  | 17     | 16 | 28.98663 |
| 406.12  | 3.88758910  | 27  | 15     | 12 | 32.68323 |
| 427.7   | 1.43512423  | 10  | 3      | 7  | 36.19020 |
| 450.3   | 3.56798827  | 19  | 13     | 6  | 36.19602 |
| 506.2   | 2.71214267  | 21  | 10     | 11 | 33.26623 |
| Canchan | 5.13246683  | 40  | 20     | 20 | 27.00126 |
| Desiree | 16.47021287 | 56  | 28     | 28 | 16.15569 |
| Unica   | 10.49672952 | 27  | 25     | 2  | 39.10400 |

```
# With n = 4 and default ssi.method (farshadfar)
ASTAB.AMMI(model, n = 4)
```

|        | ASTAB     | SSI | rASTAB | rY | means    |
|--------|-----------|-----|--------|----|----------|
| 102.18 | 4.1339139 | 36  | 13     | 23 | 26.31947 |

|         |            |    |    |    |          |
|---------|------------|----|----|----|----------|
| 104.22  | 2.3887379  | 21 | 8  | 13 | 31.28887 |
| 121.31  | 8.8192568  | 38 | 23 | 15 | 30.10174 |
| 141.28  | 7.3090299  | 22 | 21 | 1  | 39.75624 |
| 157.26  | 14.9147148 | 31 | 26 | 5  | 36.95181 |
| 163.9   | 4.8975417  | 45 | 18 | 27 | 21.41747 |
| 221.19  | 1.5353874  | 29 | 3  | 26 | 22.98480 |
| 233.11  | 2.2356017  | 24 | 7  | 17 | 28.66655 |
| 235.6   | 11.0719467 | 29 | 25 | 4  | 38.63477 |
| 241.2   | 1.7489308  | 27 | 5  | 22 | 26.34039 |
| 255.7   | 4.6032909  | 30 | 16 | 14 | 30.58975 |
| 314.12  | 2.5919840  | 27 | 9  | 18 | 28.17335 |
| 317.6   | 2.1098263  | 15 | 6  | 9  | 35.32583 |
| 319.20  | 15.5173080 | 30 | 27 | 3  | 38.75767 |
| 320.16  | 4.8783163  | 38 | 17 | 21 | 26.34808 |
| 342.15  | 4.4168665  | 39 | 15 | 24 | 26.01336 |
| 346.2   | 8.3050795  | 47 | 22 | 25 | 23.84175 |
| 351.26  | 7.1030587  | 28 | 20 | 8  | 36.11581 |
| 364.21  | 0.8834847  | 12 | 2  | 10 | 34.05974 |
| 402.7   | 0.1536666  | 20 | 1  | 19 | 27.47748 |
| 405.2   | 4.3356781  | 30 | 14 | 16 | 28.98663 |
| 406.12  | 4.0365553  | 24 | 12 | 12 | 32.68323 |
| 427.7   | 1.7169781  | 11 | 4  | 7  | 36.19020 |
| 450.3   | 3.9433912  | 17 | 11 | 6  | 36.19602 |
| 506.2   | 2.7143137  | 21 | 10 | 11 | 33.26623 |
| Canchan | 5.1384242  | 39 | 19 | 20 | 27.00126 |
| Desiree | 16.4723733 | 56 | 28 | 28 | 16.15569 |
| Unica   | 10.9110354 | 26 | 24 | 2  | 39.10400 |

```
# With default n (N') and ssi.method = "rao"
ASTAB.AMMI(model, ssi.method = "rao")
```

|        | ASTAB       | SSI        | rASTAB | rY | means    |
|--------|-------------|------------|--------|----|----------|
| 102.18 | 3.89636621  | 0.9916073  | 16     | 23 | 26.31947 |
| 104.22 | 2.19372771  | 1.2572096  | 8      | 13 | 31.28887 |
| 121.31 | 3.87988776  | 1.1154972  | 14     | 15 | 30.10174 |
| 141.28 | 7.24523520  | 1.3680406  | 22     | 1  | 39.75624 |
| 157.26 | 11.05196482 | 1.2518822  | 26     | 5  | 36.95181 |
| 163.9  | 4.64005014  | 0.8103867  | 19     | 27 | 21.41747 |
| 221.19 | 1.52227265  | 1.0909958  | 4      | 26 | 22.98480 |
| 233.11 | 2.18330553  | 1.1728390  | 7      | 17 | 28.66655 |
| 235.6  | 10.03128021 | 1.3115430  | 24     | 4  | 38.63477 |
| 241.2  | 1.65890425  | 1.1722749  | 5      | 22 | 26.34039 |
| 255.7  | 4.50083178  | 1.1129205  | 18     | 14 | 30.58975 |
| 314.12 | 2.58839912  | 1.1194868  | 9      | 18 | 28.17335 |
| 317.6  | 1.77133006  | 1.4453573  | 6      | 9  | 35.32583 |
| 319.20 | 14.26494686 | 1.3001667  | 27     | 3  | 38.75767 |
| 320.16 | 3.13335427  | 1.0250358  | 11     | 21 | 26.34808 |
| 342.15 | 3.16217247  | 1.0126098  | 12     | 24 | 26.01336 |
| 346.2  | 7.47744386  | 0.8469106  | 23     | 25 | 23.84175 |
| 351.26 | 7.10182225  | 1.2507915  | 21     | 8  | 36.11581 |
| 364.21 | 0.27632429  | 2.9922101  | 2      | 10 | 34.05974 |
| 402.7  | 0.02344768  | 23.0708927 | 1      | 19 | 27.47748 |
| 405.2  | 4.07390905  | 1.0727560  | 17     | 16 | 28.98663 |
| 406.12 | 3.88758910  | 1.1994027  | 15     | 12 | 32.68323 |
| 427.7  | 1.43512423  | 1.5423074  | 3      | 7  | 36.19020 |

```

450.3    3.56798827  1.3259199    13  6 36.19602
506.2    2.71214267  1.2763780    10 11 33.26623
Canchan  5.13246683  0.9816986    20 20 27.00126
Desiree  16.47021287  0.5583351    28 28 16.15569
Unica    10.49672952  1.3245441    25  2 39.10400

```

```

# Changing the ratio of weights for Rao's SSI
ASTAB.AMMI(model, ssi.method = "rao", a = 0.43)

```

|         | ASTAB       | SSI        | rASTAB | rY | means    |
|---------|-------------|------------|--------|----|----------|
| 102.18  | 3.89636621  | 0.9155436  | 16     | 23 | 26.31947 |
| 104.22  | 2.19372771  | 1.1221097  | 8      | 13 | 31.28887 |
| 121.31  | 3.87988776  | 1.0391104  | 14     | 15 | 30.10174 |
| 141.28  | 7.24523520  | 1.3271348  | 22     | 1  | 39.75624 |
| 157.26  | 11.05196482 | 1.2250659  | 26     | 5  | 36.95181 |
| 163.9   | 4.64005014  | 0.7465140  | 19     | 27 | 21.41747 |
| 221.19  | 1.52227265  | 0.8963051  | 4      | 26 | 22.98480 |
| 233.11  | 2.18330553  | 1.0370941  | 7      | 17 | 28.66655 |
| 235.6   | 10.03128021 | 1.2819982  | 24     | 4  | 38.63477 |
| 241.2   | 1.65890425  | 0.9936194  | 5      | 22 | 26.34039 |
| 255.7   | 4.50083178  | 1.0470721  | 18     | 14 | 30.58975 |
| 314.12  | 2.58839912  | 1.0049865  | 9      | 18 | 28.17335 |
| 317.6   | 1.77133006  | 1.2780410  | 6      | 9  | 35.32583 |
| 319.20  | 14.26494686 | 1.2793904  | 27     | 3  | 38.75767 |
| 320.16  | 3.13335427  | 0.9304495  | 11     | 21 | 26.34808 |
| 342.15  | 3.16217247  | 0.9188855  | 12     | 24 | 26.01336 |
| 346.2   | 7.47744386  | 0.8072751  | 23     | 25 | 23.84175 |
| 351.26  | 7.10182225  | 1.2090596  | 21     | 8  | 36.11581 |
| 364.21  | 0.27632429  | 1.9196572  | 2      | 10 | 34.05974 |
| 402.7   | 0.02344768  | 10.4311581 | 1      | 19 | 27.47748 |
| 405.2   | 4.07390905  | 1.0000071  | 17     | 16 | 28.98663 |
| 406.12  | 3.88758910  | 1.1231672  | 15     | 12 | 32.68323 |
| 427.7   | 1.43512423  | 1.3357940  | 3      | 7  | 36.19020 |
| 450.3   | 3.56798827  | 1.2428556  | 13     | 6  | 36.19602 |
| 506.2   | 2.71214267  | 1.1671018  | 10     | 11 | 33.26623 |
| Canchan | 5.13246683  | 0.9239540  | 20     | 20 | 27.00126 |
| Desiree | 16.47021287 | 0.5403407  | 28     | 28 | 16.15569 |
| Unica   | 10.49672952 | 1.2963093  | 25     | 2  | 39.10400 |

```

# With default n (N') and default ssi.method (farshadfar)
AVAMGE.AMMI(model)

```

```
AVAMGE.AMMI()
```

|        | AVAMGE    | SSI | rAVAMGE | rY | means    |
|--------|-----------|-----|---------|----|----------|
| 102.18 | 30.229771 | 40  | 17      | 23 | 26.31947 |
| 104.22 | 21.584579 | 21  | 8       | 13 | 31.28887 |
| 121.31 | 27.893984 | 28  | 13      | 15 | 30.10174 |
| 141.28 | 40.486706 | 24  | 23      | 1  | 39.75624 |
| 157.26 | 44.055803 | 29  | 24      | 5  | 36.95181 |
| 163.9  | 39.056228 | 48  | 21      | 27 | 21.41747 |
| 221.19 | 17.905975 | 33  | 7       | 26 | 22.98480 |
| 233.11 | 16.242635 | 21  | 4       | 17 | 28.66655 |
| 235.6  | 39.840739 | 26  | 22      | 4  | 38.63477 |
| 241.2  | 17.101113 | 28  | 6       | 22 | 26.34039 |

|         |           |    |    |    |          |
|---------|-----------|----|----|----|----------|
| 255.7   | 29.306918 | 29 | 15 | 14 | 30.58975 |
| 314.12  | 28.760304 | 32 | 14 | 18 | 28.17335 |
| 317.6   | 22.700856 | 18 | 9  | 9  | 35.32583 |
| 319.20  | 55.232023 | 30 | 27 | 3  | 38.75767 |
| 320.16  | 30.717681 | 40 | 19 | 21 | 26.34808 |
| 342.15  | 25.538281 | 34 | 10 | 24 | 26.01336 |
| 346.2   | 46.236590 | 50 | 25 | 25 | 23.84175 |
| 351.26  | 30.105573 | 24 | 16 | 8  | 36.11581 |
| 364.21  | 6.742386  | 12 | 2  | 10 | 34.05974 |
| 402.7   | 2.202291  | 20 | 1  | 19 | 27.47748 |
| 405.2   | 35.890684 | 36 | 20 | 16 | 28.98663 |
| 406.12  | 27.272847 | 24 | 12 | 12 | 32.68323 |
| 427.7   | 16.756971 | 12 | 5  | 7  | 36.19020 |
| 450.3   | 25.628188 | 17 | 11 | 6  | 36.19602 |
| 506.2   | 15.760611 | 14 | 3  | 11 | 33.26623 |
| Canchan | 30.515224 | 38 | 18 | 20 | 27.00126 |
| Desiree | 69.096357 | 56 | 28 | 28 | 16.15569 |
| Unica   | 47.204593 | 28 | 26 | 2  | 39.10400 |

```
# With n = 4 and default ssi.method (farshadfar)
AVAMGE.AMMI(model, n = 4)
```

|         | AVAMGE    | SSI | rAVAMGE | rY | means    |
|---------|-----------|-----|---------|----|----------|
| 102.18  | 30.431550 | 39  | 16      | 23 | 26.31947 |
| 104.22  | 21.176775 | 21  | 8       | 13 | 31.28887 |
| 121.31  | 34.844853 | 34  | 19      | 15 | 30.10174 |
| 141.28  | 40.382139 | 24  | 23      | 1  | 39.75624 |
| 157.26  | 49.421992 | 31  | 26      | 5  | 36.95181 |
| 163.9   | 38.846149 | 48  | 21      | 27 | 21.41747 |
| 221.19  | 17.858564 | 33  | 7       | 26 | 22.98480 |
| 233.11  | 17.449539 | 23  | 6       | 17 | 28.66655 |
| 235.6   | 39.657410 | 26  | 22      | 4  | 38.63477 |
| 241.2   | 17.225331 | 27  | 5       | 22 | 26.34039 |
| 255.7   | 29.585043 | 28  | 14      | 14 | 30.58975 |
| 314.12  | 28.801567 | 31  | 13      | 18 | 28.17335 |
| 317.6   | 23.101824 | 18  | 9       | 9  | 35.32583 |
| 319.20  | 55.695327 | 30  | 27      | 3  | 38.75767 |
| 320.16  | 31.566364 | 39  | 18      | 21 | 26.34808 |
| 342.15  | 26.310253 | 35  | 11      | 24 | 26.01336 |
| 346.2   | 46.863568 | 50  | 25      | 25 | 23.84175 |
| 351.26  | 29.920025 | 23  | 15      | 8  | 36.11581 |
| 364.21  | 9.635146  | 12  | 2       | 10 | 34.05974 |
| 402.7   | 3.665565  | 20  | 1       | 19 | 27.47748 |
| 405.2   | 35.538076 | 36  | 20      | 16 | 28.98663 |
| 406.12  | 26.916422 | 24  | 12      | 12 | 32.68323 |
| 427.7   | 16.266701 | 11  | 4       | 7  | 36.19020 |
| 450.3   | 25.622916 | 16  | 10      | 6  | 36.19602 |
| 506.2   | 15.709209 | 14  | 3       | 11 | 33.26623 |
| Canchan | 30.908627 | 37  | 17      | 20 | 27.00126 |
| Desiree | 69.115600 | 56  | 28      | 28 | 16.15569 |
| Unica   | 46.610186 | 26  | 24      | 2  | 39.10400 |

```
# With default n (N') and ssi.method = "rao"
AVAMGE.AMMI(model, ssi.method = "rao")
```



|         | AVAMGE    | SSI       | rAVAMGE | rY | means    |
|---------|-----------|-----------|---------|----|----------|
| 102.18  | 30.229771 | 1.4579240 | 17      | 23 | 26.31947 |
| 104.22  | 21.584579 | 1.8601746 | 8       | 13 | 31.28887 |
| 121.31  | 27.893984 | 1.6314700 | 13      | 15 | 30.10174 |
| 141.28  | 40.486706 | 1.7440938 | 23      | 1  | 39.75624 |
| 157.26  | 44.055803 | 1.6163747 | 24      | 5  | 36.95181 |
| 163.9   | 39.056228 | 1.1625489 | 21      | 27 | 21.41747 |
| 221.19  | 17.905975 | 1.7619814 | 7       | 26 | 22.98480 |
| 233.11  | 16.242635 | 2.0509293 | 4       | 17 | 28.66655 |
| 235.6   | 39.840739 | 1.7147885 | 22      | 4  | 38.63477 |
| 241.2   | 17.101113 | 1.9190480 | 6       | 22 | 26.34039 |
| 255.7   | 29.306918 | 1.6160450 | 15      | 14 | 30.58975 |
| 314.12  | 28.760304 | 1.5490150 | 14      | 18 | 28.17335 |
| 317.6   | 22.700856 | 1.9504975 | 9       | 9  | 35.32583 |
| 319.20  | 55.232023 | 1.5919808 | 27      | 3  | 38.75767 |
| 320.16  | 30.717681 | 1.4493304 | 19      | 21 | 26.34808 |
| 342.15  | 25.538281 | 1.5581219 | 10      | 24 | 26.01336 |
| 346.2   | 46.236590 | 1.1695027 | 25      | 25 | 23.84175 |
| 351.26  | 30.105573 | 1.7798138 | 16      | 8  | 36.11581 |
| 364.21  | 6.742386  | 3.7995961 | 2       | 10 | 34.05974 |
| 402.7   | 2.202291  | 9.1285592 | 1       | 19 | 27.47748 |
| 405.2   | 35.890684 | 1.4502899 | 20      | 16 | 28.98663 |
| 406.12  | 27.272847 | 1.7304443 | 12      | 12 | 32.68323 |
| 427.7   | 16.756971 | 2.2619806 | 5       | 7  | 36.19020 |
| 450.3   | 25.628188 | 1.8876432 | 11      | 6  | 36.19602 |
| 506.2   | 15.760611 | 2.2350438 | 3       | 11 | 33.26623 |
| Canchan | 30.515224 | 1.4745437 | 18      | 20 | 27.00126 |
| Desiree | 69.096357 | 0.7891628 | 28      | 28 | 16.15569 |
| Unica   | 47.204593 | 1.6590963 | 26      | 2  | 39.10400 |

```
# Changing the ratio of weights for Rao's SSI
AVAMGE.AMMI(model, ssi.method = "rao", a = 0.43)
```

|        | AVAMGE    | SSI       | rAVAMGE | rY | means    |
|--------|-----------|-----------|---------|----|----------|
| 102.18 | 30.229771 | 1.1160597 | 17      | 23 | 26.31947 |
| 104.22 | 21.584579 | 1.3813847 | 8       | 13 | 31.28887 |
| 121.31 | 27.893984 | 1.2609787 | 13      | 15 | 30.10174 |
| 141.28 | 40.486706 | 1.4888376 | 23      | 1  | 39.75624 |
| 157.26 | 44.055803 | 1.3817977 | 24      | 5  | 36.95181 |
| 163.9  | 39.056228 | 0.8979438 | 21      | 27 | 21.41747 |
| 221.19 | 17.905975 | 1.1848289 | 7       | 26 | 22.98480 |
| 233.11 | 16.242635 | 1.4146730 | 4       | 17 | 28.66655 |
| 235.6  | 39.840739 | 1.4553938 | 22      | 4  | 38.63477 |
| 241.2  | 17.101113 | 1.3147318 | 6       | 22 | 26.34039 |
| 255.7  | 29.306918 | 1.2634156 | 15      | 14 | 30.58975 |
| 314.12 | 28.760304 | 1.1896837 | 14      | 18 | 28.17335 |
| 317.6  | 22.700856 | 1.4952513 | 9       | 9  | 35.32583 |
| 319.20 | 55.232023 | 1.4048705 | 27      | 3  | 38.75767 |
| 320.16 | 30.717681 | 1.1128962 | 19      | 21 | 26.34808 |
| 342.15 | 25.538281 | 1.1534557 | 10      | 24 | 26.01336 |
| 346.2  | 46.236590 | 0.9459897 | 25      | 25 | 23.84175 |
| 351.26 | 30.105573 | 1.4365392 | 16      | 8  | 36.11581 |
| 364.21 | 6.742386  | 2.2668332 | 2       | 10 | 34.05974 |
| 402.7  | 2.202291  | 4.4359547 | 1       | 19 | 27.47748 |
| 405.2  | 35.890684 | 1.1623466 | 20      | 16 | 28.98663 |

|         |           |           |    |    |          |
|---------|-----------|-----------|----|----|----------|
| 406.12  | 27.272847 | 1.3515151 | 12 | 12 | 32.68323 |
| 427.7   | 16.756971 | 1.6452535 | 5  | 7  | 36.19020 |
| 450.3   | 25.628188 | 1.4843966 | 11 | 6  | 36.19602 |
| 506.2   | 15.760611 | 1.5793281 | 3  | 11 | 33.26623 |
| Canchan | 30.515224 | 1.1358773 | 18 | 20 | 27.00126 |
| Desiree | 69.096357 | 0.6395966 | 28 | 28 | 16.15569 |
| Unica   | 47.204593 | 1.4401668 | 26 | 2  | 39.10400 |

```
# With default n (N') and default ssi.method (farshadfar)
DA.AMMI(model)
```

```
DA.AMMI()
```

|         | DA        | SSI | rDA | rY | means    |
|---------|-----------|-----|-----|----|----------|
| 102.18  | 15.040431 | 39  | 16  | 23 | 26.31947 |
| 104.22  | 9.798867  | 22  | 9   | 13 | 31.28887 |
| 121.31  | 12.917859 | 26  | 11  | 15 | 30.10174 |
| 141.28  | 19.659222 | 23  | 22  | 1  | 39.75624 |
| 157.26  | 21.459064 | 29  | 24  | 5  | 36.95181 |
| 163.9   | 17.499098 | 48  | 21  | 27 | 21.41747 |
| 221.19  | 8.507426  | 31  | 5   | 26 | 22.98480 |
| 233.11  | 8.981297  | 24  | 7   | 17 | 28.66655 |
| 235.6   | 21.941275 | 29  | 25  | 4  | 38.63477 |
| 241.2   | 8.453875  | 26  | 4   | 22 | 26.34039 |
| 255.7   | 15.423064 | 32  | 18  | 14 | 30.58975 |
| 314.12  | 12.222308 | 28  | 10  | 18 | 28.17335 |
| 317.6   | 9.592839  | 17  | 8   | 9  | 35.32583 |
| 319.20  | 28.986374 | 30  | 27  | 3  | 38.75767 |
| 320.16  | 13.835583 | 34  | 13  | 21 | 26.34808 |
| 342.15  | 13.025230 | 36  | 12  | 24 | 26.01336 |
| 346.2   | 21.230207 | 48  | 23  | 25 | 23.84175 |
| 351.26  | 17.269543 | 28  | 20  | 8  | 36.11581 |
| 364.21  | 3.781576  | 12  | 2   | 10 | 34.05974 |
| 402.7   | 1.191312  | 20  | 1   | 19 | 27.47748 |
| 405.2   | 16.027557 | 35  | 19  | 16 | 28.98663 |
| 406.12  | 13.989359 | 26  | 14  | 12 | 32.68323 |
| 427.7   | 7.507408  | 10  | 3   | 7  | 36.19020 |
| 450.3   | 14.270920 | 21  | 15  | 6  | 36.19602 |
| 506.2   | 8.954538  | 17  | 6   | 11 | 33.26623 |
| Canchan | 15.138085 | 37  | 17  | 20 | 27.00126 |
| Desiree | 32.114860 | 56  | 28  | 28 | 16.15569 |
| Unica   | 22.343936 | 28  | 26  | 2  | 39.10400 |

```
# With n = 4 and default ssi.method (farshadfar)
DA.AMMI(model, n = 4)
```

|        | DA        | SSI | rDA | rY | means    |
|--------|-----------|-----|-----|----|----------|
| 102.18 | 15.185880 | 39  | 16  | 23 | 26.31947 |
| 104.22 | 9.981329  | 22  | 9   | 13 | 31.28887 |
| 121.31 | 16.071287 | 33  | 18  | 15 | 30.10174 |
| 141.28 | 19.689228 | 23  | 22  | 1  | 39.75624 |
| 157.26 | 23.064716 | 31  | 26  | 5  | 36.95181 |
| 163.9  | 17.634737 | 48  | 21  | 27 | 21.41747 |
| 221.19 | 8.521680  | 30  | 4   | 26 | 22.98480 |
| 233.11 | 9.035019  | 24  | 7   | 17 | 28.66655 |

```

235.6  22.375871  28  24  4 38.63477
241.2   8.551852  27   5 22 26.34039
255.7  15.484417  31  17 14 30.58975
314.12 12.225021  28  10 18 28.17335
317.6   9.913993  17   8  9 35.32583
319.20 29.383463  30  27  3 38.75767
320.16 14.957211  35  14 21 26.34808
342.15 13.888046  35  11 24 26.01336
346.2   21.587939  48  23 25 23.84175
351.26 17.270205  28  20  8 36.11581
364.21  5.053446  12   2 10 34.05974
402.7   1.956846  20   1 19 27.47748
405.2   16.177987  35  19 16 28.98663
406.12 14.087553  24  12 12 32.68323
427.7   7.847138  10   3  7 36.19020
450.3   14.512302  19  13  6 36.19602
506.2   8.956781  17   6 11 33.26623
Canchan 15.141726  35  15 20 27.00126
Desiree 32.115482  56  28 28 16.15569
Unica   22.514867  27  25  2 39.10400

```

```

# With default n (N') and ssi.method = "rao"
DA.AMMI(model, ssi.method = "rao")

```

```

          DA          SSI rDA rY    means
102.18 15.040431 1.4730947 16 23 26.31947
104.22  9.798867 1.9640618  9 13 31.28887
121.31 12.917859 1.6974593 11 15 30.10174
141.28 19.659222 1.7667347 22  1 39.75624
157.26 21.459064 1.6358359 24  5 36.95181
163.9  17.499098 1.2268624 21 27 21.41747
221.19  8.507426 1.8365835  5 26 22.98480
233.11  8.981297 1.9644804  7 17 28.66655
235.6  21.941275 1.6812376 25  4 38.63477
241.2   8.453875 1.9528811  4 22 26.34039
255.7  15.423064 1.5970737 18 14 30.58975
314.12 12.222308 1.6753281 10 18 28.17335
317.6   9.592839 2.1159612  8  9 35.32583
319.20 28.986374 1.5827930 27  3 38.75767
320.16 13.835583 1.5275780 13 21 26.34808
342.15 13.025230 1.5582533 12 24 26.01336
346.2   21.230207 1.2130205 23 25 23.84175
351.26 17.269543 1.7131362 20  8 36.11581
364.21  3.781576 3.5563052  2 10 34.05974
402.7   1.191312 8.6595018  1 19 27.47748
405.2   16.027557 1.5221857 19 16 28.98663
406.12 13.989359 1.7267910 14 12 32.68323
427.7   7.507408 2.4119665  3  7 36.19020
450.3   14.270920 1.8282838 15  6 36.19602
506.2   8.954538 2.1175331  6 11 33.26623
Canchan 15.138085 1.4913580 17 20 27.00126
Desiree 32.114860 0.8147588 28 28 16.15569
Unica   22.343936 1.6889406 26  2 39.10400

```

```
# Changing the ratio of weights for Rao's SSI
DA.AMMI(model, ssi.method = "rao", a = 0.43)
```

|         | DA        | SSI       | rDA | rY | means    |
|---------|-----------|-----------|-----|----|----------|
| 102.18  | 15.040431 | 1.1225831 | 16  | 23 | 26.31947 |
| 104.22  | 9.798867  | 1.4260562 | 9   | 13 | 31.28887 |
| 121.31  | 12.917859 | 1.2893541 | 11  | 15 | 30.10174 |
| 141.28  | 19.659222 | 1.4985733 | 22  | 1  | 39.75624 |
| 157.26  | 21.459064 | 1.3901660 | 24  | 5  | 36.95181 |
| 163.9   | 17.499098 | 0.9255986 | 21  | 27 | 21.41747 |
| 221.19  | 8.507426  | 1.2169078 | 5   | 26 | 22.98480 |
| 233.11  | 8.981297  | 1.3775000 | 7   | 17 | 28.66655 |
| 235.6   | 21.941275 | 1.4409668 | 25  | 4  | 38.63477 |
| 241.2   | 8.453875  | 1.3292801 | 4   | 22 | 26.34039 |
| 255.7   | 15.423064 | 1.2552580 | 18  | 14 | 30.58975 |
| 314.12  | 12.222308 | 1.2439983 | 10  | 18 | 28.17335 |
| 317.6   | 9.592839  | 1.5664007 | 8   | 9  | 35.32583 |
| 319.20  | 28.986374 | 1.4009197 | 27  | 3  | 38.75767 |
| 320.16  | 13.835583 | 1.1465427 | 13  | 21 | 26.34808 |
| 342.15  | 13.025230 | 1.1535122 | 12  | 24 | 26.01336 |
| 346.2   | 21.230207 | 0.9647024 | 23  | 25 | 23.84175 |
| 351.26  | 17.269543 | 1.4078678 | 20  | 8  | 36.11581 |
| 364.21  | 3.781576  | 2.1622181 | 2   | 10 | 34.05974 |
| 402.7   | 1.191312  | 4.2342600 | 1   | 19 | 27.47748 |
| 405.2   | 16.027557 | 1.1932619 | 19  | 16 | 28.98663 |
| 406.12  | 13.989359 | 1.3499442 | 14  | 12 | 32.68323 |
| 427.7   | 7.507408  | 1.7097474 | 3   | 7  | 36.19020 |
| 450.3   | 14.270920 | 1.4588721 | 15  | 6  | 36.19602 |
| 506.2   | 8.954538  | 1.5287986 | 6   | 11 | 33.26623 |
| Canchan | 15.138085 | 1.1431075 | 17  | 20 | 27.00126 |
| Desiree | 32.114860 | 0.6506029 | 28  | 28 | 16.15569 |
| Unica   | 22.343936 | 1.4529998 | 26  | 2  | 39.10400 |

```
# With default n (N') and default ssi.method (farshadfar)
DZ.AMMI(model)
```

```
DZ.AMMI()
```

|        | DZ         | SSI | rDZ | rY | means    |
|--------|------------|-----|-----|----|----------|
| 102.18 | 0.26393535 | 37  | 14  | 23 | 26.31947 |
| 104.22 | 0.22971564 | 21  | 8   | 13 | 31.28887 |
| 121.31 | 0.32031744 | 34  | 19  | 15 | 30.10174 |
| 141.28 | 0.39838535 | 23  | 22  | 1  | 39.75624 |
| 157.26 | 0.53822924 | 33  | 28  | 5  | 36.95181 |
| 163.9  | 0.26659011 | 42  | 15  | 27 | 21.41747 |
| 221.19 | 0.19563325 | 29  | 3   | 26 | 22.98480 |
| 233.11 | 0.25167755 | 27  | 10  | 17 | 28.66655 |
| 235.6  | 0.46581370 | 28  | 24  | 4  | 38.63477 |
| 241.2  | 0.21481887 | 28  | 6   | 22 | 26.34039 |
| 255.7  | 0.30862904 | 31  | 17  | 14 | 30.58975 |
| 314.12 | 0.22603261 | 25  | 7   | 18 | 28.17335 |
| 317.6  | 0.20224771 | 14  | 5   | 9  | 35.32583 |
| 319.20 | 0.50675112 | 29  | 26  | 3  | 38.75767 |
| 320.16 | 0.23280596 | 30  | 9   | 21 | 26.34808 |

```

342.15  0.25989774  36  12  24  26.01336
346.2   0.37125512  45  20  25  23.84175
351.26  0.43805896  31  23   8  36.11581
364.21  0.07409309  12   2  10  34.05974
402.7   0.02004533  20   1  19  27.47748
405.2   0.26238837  29  13  16  28.98663
406.12  0.28179394  28  16  12  32.68323
427.7   0.20176581  11   4   7  36.19020
450.3   0.25465368  17  11   6  36.19602
506.2   0.30899851  29  18  11  33.26623
Canchan 0.37201039  41  21  20  27.00126
Desiree 0.52005815  55  27  28  16.15569
Unica   0.48083049  27  25   2  39.10400

```

```

# With n = 4 and default ssi.method (farshadfar)
DZ.AMMI(model, n = 4)

```

```

          DZ SSI rDZ rY    means
102.18  0.28722309  33  10  23  26.31947
104.22  0.25160706  21   8  13  31.28887
121.31  0.60785568  42  27  15  30.10174
141.28  0.40268829  21  20   1  39.75624
157.26  0.70597721  33  28   5  36.95181
163.9   0.29151868  39  12  27  21.41747
221.19  0.19743603  29   3  26  22.98480
233.11  0.25722999  26   9  17  28.66655
235.6   0.52269682  29  25   4  38.63477
241.2   0.22585722  26   4  22  26.34039
255.7   0.31747123  30  16  14  30.58975
314.12  0.22646067  23   5  18  28.17335
317.6   0.24329787  16   7   9  35.32583
319.20  0.56961794  29  26   3  38.75767
320.16  0.38533472  40  19  21  26.34808
342.15  0.36788692  41  17  24  26.01336
346.2   0.42725798  46  21  25  23.84175
351.26  0.43813521  30  22   8  36.11581
364.21  0.19569373  12   2  10  34.05974
402.7   0.08624291  20   1  19  27.47748
405.2   0.28808268  27  11  16  28.98663
406.12  0.29573097  26  14  12  32.68323
427.7   0.23651352  13   6   7  36.19020
450.3   0.29177451  19  13   6  36.19602
506.2   0.30918827  26  15  11  33.26623
Canchan 0.37244277  38  18  20  27.00126
Desiree 0.52017037  52  24  28  16.15569
Unica   0.50357109  25  23   2  39.10400

```

```

# With default n (N') and ssi.method = "rao"
DZ.AMMI(model, ssi.method = "rao")

```

```

          DZ          SSI rDZ rY    means
102.18  0.26393535  1.5536988  14  23  26.31947
104.22  0.22971564  1.8193399   8  13  31.28887
121.31  0.32031744  1.5545939  19  15  30.10174
141.28  0.39838535  1.7570779  22   1  39.75624

```

```

157.26  0.53822924  1.5459114  28  5 36.95181
163.9   0.26659011  1.3869397  15 27 21.41747
221.19  0.19563325  1.6878048   3 26 22.98480
233.11  0.25167755  1.6641025  10 17 28.66655
235.6   0.46581370  1.6538090  24  4 38.63477
241.2   0.21481887  1.7134093   6 22 26.34039
255.7   0.30862904  1.5922105  17 14 30.58975
314.12  0.22603261  1.7307783   7 18 28.17335
317.6   0.20224771  2.0595024   5  9 35.32583
319.20  0.50675112  1.6259792  26  3 38.75767
320.16  0.23280596  1.6476346   9 21 26.34808
342.15  0.25989774  1.5545233  12 24 26.01336
346.2   0.37125512  1.2718506  20 25 23.84175
351.26  0.43805896  1.5966462  23  8 36.11581
364.21  0.07409309  3.5881882   2 10 34.05974
402.7   0.02004533 10.0539968   1 19 27.47748
405.2   0.26238837  1.6447637  13 16 28.98663
406.12  0.28179394  1.7171135  16 12 32.68323
427.7   0.20176581  2.0898536   4  7 36.19020
450.3   0.25465368  1.9010808  11  6 36.19602
506.2   0.30899851  1.6787677  18 11 33.26623
Canchan 0.37201039  1.3738642  21 20 27.00126
Desiree 0.52005815  0.8797586  27 28 16.15569
Unica   0.48083049  1.6568004  25  2 39.10400

```

```

# Changing the ratio of weights for Rao's SSI
DZ.AMMI(model, ssi.method = "rao", a = 0.43)

```

```

          DZ      SSI rDZ rY    means
102.18  0.26393535  1.1572429  14 23 26.31947
104.22  0.22971564  1.3638258   8 13 31.28887
121.31  0.32031744  1.2279220  19 15 30.10174
141.28  0.39838535  1.4944208  22  1 39.75624
157.26  0.53822924  1.3514985  28  5 36.95181
163.9   0.26659011  0.9944318  15 27 21.41747
221.19  0.19563325  1.1529329   3 26 22.98480
233.11  0.25167755  1.2483375  10 17 28.66655
235.6   0.46581370  1.4291726  24  4 38.63477
241.2   0.21481887  1.2263072   6 22 26.34039
255.7   0.30862904  1.2531668  17 14 30.58975
314.12  0.22603261  1.2678419   7 18 28.17335
317.6   0.20224771  1.5421234   5  9 35.32583
319.20  0.50675112  1.4194898  26  3 38.75767
320.16  0.23280596  1.1981670   9 21 26.34808
342.15  0.25989774  1.1519083  12 24 26.01336
346.2   0.37125512  0.9899993  20 25 23.84175
351.26  0.43805896  1.3577771  23  8 36.11581
364.21  0.07409309  2.1759278   2 10 34.05974
402.7   0.02004533  4.8338929   1 19 27.47748
405.2   0.26238837  1.2459704  13 16 28.98663
406.12  0.28179394  1.3457828  16 12 32.68323
427.7   0.20176581  1.5712389   4  7 36.19020
450.3   0.25465368  1.4901748  11  6 36.19602
506.2   0.30899851  1.3401295  18 11 33.26623
Canchan 0.37201039  1.0925852  21 20 27.00126

```

```
Desiree 0.52005815 0.6785528 27 28 16.15569
Unica 0.48083049 1.4391795 25 2 39.10400
```

```
# With default n (N') and default ssi.method (farshadfar)
EV.AMMI(model)
```

```
EV.AMMI()
```

|         | EV           | SSI | rEV | rY | means    |
|---------|--------------|-----|-----|----|----------|
| 102.18  | 0.0232206231 | 37  | 14  | 23 | 26.31947 |
| 104.22  | 0.0175897578 | 21  | 8   | 13 | 31.28887 |
| 121.31  | 0.0342010876 | 34  | 19  | 15 | 30.10174 |
| 141.28  | 0.0529036285 | 23  | 22  | 1  | 39.75624 |
| 157.26  | 0.0965635719 | 33  | 28  | 5  | 36.95181 |
| 163.9   | 0.0236900961 | 42  | 15  | 27 | 21.41747 |
| 221.19  | 0.0127574566 | 29  | 3   | 26 | 22.98480 |
| 233.11  | 0.0211138628 | 27  | 10  | 17 | 28.66655 |
| 235.6   | 0.0723274691 | 28  | 24  | 4  | 38.63477 |
| 241.2   | 0.0153823821 | 28  | 6   | 22 | 26.34039 |
| 255.7   | 0.0317506280 | 31  | 17  | 14 | 30.58975 |
| 314.12  | 0.0170302467 | 25  | 7   | 18 | 28.17335 |
| 317.6   | 0.0136347120 | 14  | 5   | 9  | 35.32583 |
| 319.20  | 0.0855988994 | 29  | 26  | 3  | 38.75767 |
| 320.16  | 0.0180662044 | 30  | 9   | 21 | 26.34808 |
| 342.15  | 0.0225156118 | 36  | 12  | 24 | 26.01336 |
| 346.2   | 0.0459434537 | 45  | 20  | 25 | 23.84175 |
| 351.26  | 0.0639652186 | 31  | 23  | 8  | 36.11581 |
| 364.21  | 0.0018299284 | 12  | 2   | 10 | 34.05974 |
| 402.7   | 0.0001339385 | 20  | 1   | 19 | 27.47748 |
| 405.2   | 0.0229492190 | 29  | 13  | 16 | 28.98663 |
| 406.12  | 0.0264692745 | 28  | 16  | 12 | 32.68323 |
| 427.7   | 0.0135698145 | 11  | 4   | 7  | 36.19020 |
| 450.3   | 0.0216161656 | 17  | 11  | 6  | 36.19602 |
| 506.2   | 0.0318266934 | 29  | 18  | 11 | 33.26623 |
| Canchan | 0.0461305761 | 41  | 21  | 20 | 27.00126 |
| Desiree | 0.0901534938 | 55  | 27  | 28 | 16.15569 |
| Unica   | 0.0770659860 | 27  | 25  | 2  | 39.10400 |

```
# With n = 4 and default ssi.method (farshadfar)
EV.AMMI(model, n = 4)
```

|        | EV          | SSI | rEV | rY | means    |
|--------|-------------|-----|-----|----|----------|
| 102.18 | 0.020624276 | 33  | 10  | 23 | 26.31947 |
| 104.22 | 0.015826528 | 21  | 8   | 13 | 31.28887 |
| 121.31 | 0.092372131 | 42  | 27  | 15 | 30.10174 |
| 141.28 | 0.040539465 | 21  | 20  | 1  | 39.75624 |
| 157.26 | 0.124600955 | 33  | 28  | 5  | 36.95181 |
| 163.9  | 0.021245785 | 39  | 12  | 27 | 21.41747 |
| 221.19 | 0.009745247 | 29  | 3   | 26 | 22.98480 |
| 233.11 | 0.016541818 | 26  | 9   | 17 | 28.66655 |
| 235.6  | 0.068302992 | 29  | 25  | 4  | 38.63477 |
| 241.2  | 0.012752871 | 26  | 4   | 22 | 26.34039 |
| 255.7  | 0.025196996 | 30  | 16  | 14 | 30.58975 |
| 314.12 | 0.012821109 | 23  | 5   | 18 | 28.17335 |
| 317.6  | 0.014798464 | 16  | 7   | 9  | 35.32583 |

```

319.20 0.081116150 29 26 3 38.75767
320.16 0.037120712 40 19 21 26.34808
342.15 0.033835196 41 17 24 26.01336
346.2 0.045637346 46 21 25 23.84175
351.26 0.047990616 30 22 8 36.11581
364.21 0.009574009 12 2 10 34.05974
402.7 0.001859460 20 1 19 27.47748
405.2 0.020747907 27 11 16 28.98663
406.12 0.021864201 26 14 12 32.68323
427.7 0.013984661 13 6 7 36.19020
450.3 0.021283092 19 13 6 36.19602
506.2 0.023899346 26 15 11 33.26623
Canchan 0.034678404 38 18 20 27.00126
Desiree 0.067644303 52 24 28 16.15569
Unica 0.063395960 25 23 2 39.10400

```

```

# With default n (N') and ssi.method = "rao"
EV.AMMI(model, ssi.method = "rao")

```

|         | EV           | SSI        | rEV | rY | means    |
|---------|--------------|------------|-----|----|----------|
| 102.18  | 0.0232206231 | 0.9920136  | 14  | 23 | 26.31947 |
| 104.22  | 0.0175897578 | 1.1968926  | 8   | 13 | 31.28887 |
| 121.31  | 0.0342010876 | 1.0723629  | 19  | 15 | 30.10174 |
| 141.28  | 0.0529036285 | 1.3550266  | 22  | 1  | 39.75624 |
| 157.26  | 0.0965635719 | 1.2370234  | 28  | 5  | 36.95181 |
| 163.9   | 0.0236900961 | 0.8295284  | 15  | 27 | 21.41747 |
| 221.19  | 0.0127574566 | 0.9930645  | 3   | 26 | 22.98480 |
| 233.11  | 0.0211138628 | 1.0818975  | 10  | 17 | 28.66655 |
| 235.6   | 0.0723274691 | 1.3026828  | 24  | 4  | 38.63477 |
| 241.2   | 0.0153823821 | 1.0609011  | 6   | 22 | 26.34039 |
| 255.7   | 0.0317506280 | 1.0952885  | 17  | 14 | 30.58975 |
| 314.12  | 0.0170302467 | 1.1011148  | 7   | 18 | 28.17335 |
| 317.6   | 0.0136347120 | 1.3797760  | 5   | 9  | 35.32583 |
| 319.20  | 0.0855988994 | 1.3000274  | 26  | 3  | 38.75767 |
| 320.16  | 0.0180662044 | 1.0311353  | 9   | 21 | 26.34808 |
| 342.15  | 0.0225156118 | 0.9862240  | 12  | 24 | 26.01336 |
| 346.2   | 0.0459434537 | 0.8450255  | 20  | 25 | 23.84175 |
| 351.26  | 0.0639652186 | 1.2261684  | 23  | 8  | 36.11581 |
| 364.21  | 0.0018299284 | 2.8090292  | 2   | 10 | 34.05974 |
| 402.7   | 0.0001339385 | 24.1014741 | 1   | 19 | 27.47748 |
| 405.2   | 0.0229492190 | 1.0805609  | 13  | 16 | 28.98663 |
| 406.12  | 0.0264692745 | 1.1830798  | 16  | 12 | 32.68323 |
| 427.7   | 0.0135698145 | 1.4090495  | 4   | 7  | 36.19020 |
| 450.3   | 0.0216161656 | 1.3239797  | 11  | 6  | 36.19602 |
| 506.2   | 0.0318266934 | 1.1823230  | 18  | 11 | 33.26623 |
| Canchan | 0.0461305761 | 0.9477687  | 21  | 20 | 27.00126 |
| Desiree | 0.0901534938 | 0.5612418  | 27  | 28 | 16.15569 |
| Unica   | 0.0770659860 | 1.3153400  | 25  | 2  | 39.10400 |

```

# Changing the ratio of weights for Rao's SSI
EV.AMMI(model, ssi.method = "rao", a = 0.43)

```

|        | EV           | SSI       | rEV | rY | means    |
|--------|--------------|-----------|-----|----|----------|
| 102.18 | 0.0232206231 | 0.9157183 | 14  | 23 | 26.31947 |
| 104.22 | 0.0175897578 | 1.0961734 | 8   | 13 | 31.28887 |
| 121.31 | 0.0342010876 | 1.0205626 | 19  | 15 | 30.10174 |



|         |              |            |    |    |          |
|---------|--------------|------------|----|----|----------|
| 141.28  | 0.0529036285 | 1.3215387  | 22 | 1  | 39.75624 |
| 157.26  | 0.0965635719 | 1.2186766  | 28 | 5  | 36.95181 |
| 163.9   | 0.0236900961 | 0.7547449  | 15 | 27 | 21.41747 |
| 221.19  | 0.0127574566 | 0.8541946  | 3  | 26 | 22.98480 |
| 233.11  | 0.0211138628 | 0.9979893  | 10 | 17 | 28.66655 |
| 235.6   | 0.0723274691 | 1.2781883  | 24 | 4  | 38.63477 |
| 241.2   | 0.0153823821 | 0.9457286  | 6  | 22 | 26.34039 |
| 255.7   | 0.0317506280 | 1.0394903  | 17 | 14 | 30.58975 |
| 314.12  | 0.0170302467 | 0.9970866  | 7  | 18 | 28.17335 |
| 317.6   | 0.0136347120 | 1.2498410  | 5  | 9  | 35.32583 |
| 319.20  | 0.0855988994 | 1.2793305  | 26 | 3  | 38.75767 |
| 320.16  | 0.0180662044 | 0.9330723  | 9  | 21 | 26.34808 |
| 342.15  | 0.0225156118 | 0.9075396  | 12 | 24 | 26.01336 |
| 346.2   | 0.0459434537 | 0.8064645  | 20 | 25 | 23.84175 |
| 351.26  | 0.0639652186 | 1.1984717  | 23 | 8  | 36.11581 |
| 364.21  | 0.0018299284 | 1.8408895  | 2  | 10 | 34.05974 |
| 402.7   | 0.0001339385 | 10.8743081 | 1  | 19 | 27.47748 |
| 405.2   | 0.0229492190 | 1.0033632  | 13 | 16 | 28.98663 |
| 406.12  | 0.0264692745 | 1.1161483  | 16 | 12 | 32.68323 |
| 427.7   | 0.0135698145 | 1.2784931  | 4  | 7  | 36.19020 |
| 450.3   | 0.0216161656 | 1.2420213  | 11 | 6  | 36.19602 |
| 506.2   | 0.0318266934 | 1.1266582  | 18 | 11 | 33.26623 |
| Canchan | 0.0461305761 | 0.9093641  | 21 | 20 | 27.00126 |
| Desiree | 0.0901534938 | 0.5415905  | 27 | 28 | 16.15569 |
| Unica   | 0.0770659860 | 1.2923516  | 25 | 2  | 39.10400 |

```
# With default n (N') and default ssi.method (farshadfar)
FA.AMMI(model)
```

```
FA.AMMI()
```

|        | FA         | SSI | rFA | rY | means    |
|--------|------------|-----|-----|----|----------|
| 102.18 | 226.214559 | 39  | 16  | 23 | 26.31947 |
| 104.22 | 96.017789  | 22  | 9   | 13 | 31.28887 |
| 121.31 | 166.871081 | 26  | 11  | 15 | 30.10174 |
| 141.28 | 386.485026 | 23  | 22  | 1  | 39.75624 |
| 157.26 | 460.491413 | 29  | 24  | 5  | 36.95181 |
| 163.9  | 306.218437 | 48  | 21  | 27 | 21.41747 |
| 221.19 | 72.376305  | 31  | 5   | 26 | 22.98480 |
| 233.11 | 80.663694  | 24  | 7   | 17 | 28.66655 |
| 235.6  | 481.419528 | 29  | 25  | 4  | 38.63477 |
| 241.2  | 71.468008  | 26  | 4   | 22 | 26.34039 |
| 255.7  | 237.870912 | 32  | 18  | 14 | 30.58975 |
| 314.12 | 149.384801 | 28  | 10  | 18 | 28.17335 |
| 317.6  | 92.022551  | 17  | 8   | 9  | 35.32583 |
| 319.20 | 840.209886 | 30  | 27  | 3  | 38.75767 |
| 320.16 | 191.423345 | 34  | 13  | 21 | 26.34808 |
| 342.15 | 169.656627 | 36  | 12  | 24 | 26.01336 |
| 346.2  | 450.721670 | 48  | 23  | 25 | 23.84175 |
| 351.26 | 298.237108 | 28  | 20  | 8  | 36.11581 |
| 364.21 | 14.300314  | 12  | 2   | 10 | 34.05974 |
| 402.7  | 1.419225   | 20  | 1   | 19 | 27.47748 |
| 405.2  | 256.882577 | 35  | 19  | 16 | 28.98663 |
| 406.12 | 195.702153 | 26  | 14  | 12 | 32.68323 |

```

427.7      56.361179  10   3   7 36.19020
450.3     203.659148  21  15   6 36.19602
506.2      80.183743  17   6  11 33.26623
Canchan   229.161607  37  17  20 27.00126
Desiree 1031.364210  56  28  28 16.15569
Unica     499.251489  28  26   2 39.10400

```

```

# With n = 4 and default ssi.method (farshadfar)
FA.AMMI(model, n = 4)

```

```

          FA SSI rFA rY      means
102.18   230.610963 39  16 23 26.31947
104.22    99.626933 22   9 13 31.28887
121.31   258.286270 33  18 15 30.10174
141.28   387.665704 23  22   1 39.75624
157.26   531.981114 31  26   5 36.95181
163.9    310.983953 48  21 27 21.41747
221.19    72.619025 30   4 26 22.98480
233.11    81.631564 24   7 17 28.66655
235.6    500.679624 28  24   4 38.63477
241.2     73.134171 27   5 22 26.34039
255.7    239.767170 31  17 14 30.58975
314.12   149.451148 28  10 18 28.17335
317.6     98.287259 17   8   9 35.32583
319.20   863.387913 30  27   3 38.75767
320.16   223.718164 35  14 21 26.34808
342.15   192.877830 35  11 24 26.01336
346.2    466.039106 48  23 25 23.84175
351.26   298.259992 28  20   8 36.11581
364.21    25.537314 12   2 10 34.05974
402.7      3.829248 20   1 19 27.47748
405.2    261.727258 35  19 16 28.98663
406.12   198.459140 24  12 12 32.68323
427.7     61.577580 10   3   7 36.19020
450.3    210.606905 19  13   6 36.19602
506.2     80.223923 17   6  11 33.26623
Canchan   229.271862 35  15  20 27.00126
Desiree 1031.404193 56  28  28 16.15569
Unica     506.919240 27  25   2 39.10400

```

```

# With default n (N') and ssi.method = "rao"
FA.AMMI(model, ssi.method = "rao")

```

```

          FA          SSI rFA rY      means
102.18   226.214559 0.9902913  16 23 26.31947
104.22    96.017789 1.3314840   9 13 31.28887
121.31   166.871081 1.1606028  11 15 30.10174
141.28   386.485026 1.3736129  22   1 39.75624
157.26   460.491413 1.2697440  24   5 36.95181
163.9    306.218437 0.7959379  21 27 21.41747
221.19    72.376305 1.1624072   5 26 22.98480
233.11    80.663694 1.3052353   7 17 28.66655
235.6    481.419528 1.3217963  25   4 38.63477
241.2     71.468008 1.2770668   4 22 26.34039
255.7    237.870912 1.1230515  18 14 30.58975

```

```

314.12  149.384801  1.1186933  10 18 28.17335
317.6   92.022551  1.4766266   8  9 35.32583
319.20  840.209886  1.2992910  27  3 38.75767
320.16  191.423345  1.0152386  13 21 26.34808
342.15  169.656627  1.0243579  12 24 26.01336
346.2   450.721670  0.8436895  23 25 23.84175
351.26  298.237108  1.2777984  20  8 36.11581
364.21   14.300314  3.2006702   2 10 34.05974
402.7    1.419225 21.9563817   1 19 27.47748
405.2   256.882577  1.0614812  19 16 28.98663
406.12  195.702153  1.2183859  14 12 32.68323
427.7    56.361179  1.7103246   3  7 36.19020
450.3   203.659148  1.3269556  15  6 36.19602
506.2    80.183743  1.4574286   6 11 33.26623
Canchan 229.161607  1.0108222  17 20 27.00126
Desiree 1031.364210 0.5557465  28 28 16.15569
Unica   499.251489  1.3348781  26  2 39.10400

```

```

# Changing the ratio of weights for Rao's SSI
FA.AMMI(model, ssi.method = "rao", a = 0.43)

```

```

          FA      SSI rFA rY  means
102.18  226.214559 0.9149776 16 23 26.31947
104.22   96.017789 1.1540477  9 13 31.28887
121.31  166.871081 1.0585058 11 15 30.10174
141.28  386.485026 1.3295309 22  1 39.75624
157.26  460.491413 1.2327465 24  5 36.95181
163.9   306.218437 0.7403010 21 27 21.41747
221.19   72.376305 0.9270120  5 26 22.98480
233.11   80.663694 1.0940246  7 17 28.66655
235.6   481.419528 1.2864071 25  4 38.63477
241.2    71.468008 1.0386799  4 22 26.34039
255.7   237.870912 1.0514284 18 14 30.58975
314.12  149.384801 1.0046453 10 18 28.17335
317.6    92.022551 1.2914868   8  9 35.32583
319.20  840.209886 1.2790139 27  3 38.75767
320.16  191.423345 0.9262367 13 21 26.34808
342.15  169.656627 0.9239372 12 24 26.01336
346.2   450.721670 0.8058900 23 25 23.84175
351.26  298.237108 1.2206726 20  8 36.11581
364.21   14.300314 2.0092951  2 10 34.05974
402.7    1.419225 9.9519184   1 19 27.47748
405.2   256.882577 0.9951589 19 16 28.98663
406.12  195.702153 1.1313300 14 12 32.68323
427.7    56.361179 1.4080414   3  7 36.19020
450.3   203.659148 1.2433009 15  6 36.19602
506.2    80.183743 1.2449536   6 11 33.26623
Canchan 229.161607 0.9364771 17 20 27.00126
Desiree 1031.364210 0.5392276 28 28 16.15569
Unica   499.251489 1.3007530 26  2 39.10400

```

```

# With default n (N') and default ssi.method (farshadfar)
MASV.AMMI(model)

```

**MASV.AMMI()**

|         | MASV      | SSI | rMASV | rY | means    |
|---------|-----------|-----|-------|----|----------|
| 102.18  | 4.7855876 | 42  | 19    | 23 | 26.31947 |
| 104.22  | 3.8328358 | 25  | 12    | 13 | 31.28887 |
| 121.31  | 4.0446758 | 29  | 14    | 15 | 30.10174 |
| 141.28  | 5.1867706 | 21  | 20    | 1  | 39.75624 |
| 157.26  | 7.6459224 | 29  | 24    | 5  | 36.95181 |
| 163.9   | 4.4977055 | 43  | 16    | 27 | 21.41747 |
| 221.19  | 2.1905344 | 31  | 5     | 26 | 22.98480 |
| 233.11  | 3.1794345 | 26  | 9     | 17 | 28.66655 |
| 235.6   | 8.4913020 | 29  | 25    | 4  | 38.63477 |
| 241.2   | 2.0338659 | 26  | 4     | 22 | 26.34039 |
| 255.7   | 4.7013868 | 32  | 18    | 14 | 30.58975 |
| 314.12  | 3.1376678 | 26  | 8     | 18 | 28.17335 |
| 317.6   | 2.3345492 | 15  | 6     | 9  | 35.32583 |
| 319.20  | 8.6398087 | 30  | 27    | 3  | 38.75767 |
| 320.16  | 3.8822326 | 34  | 13    | 21 | 26.34808 |
| 342.15  | 3.6438425 | 34  | 10    | 24 | 26.01336 |
| 346.2   | 5.3987165 | 47  | 22    | 25 | 23.84175 |
| 351.26  | 5.4005468 | 31  | 23    | 8  | 36.11581 |
| 364.21  | 1.4047546 | 12  | 2     | 10 | 34.05974 |
| 402.7   | 0.3537818 | 20  | 1     | 19 | 27.47748 |
| 405.2   | 4.1095727 | 31  | 15    | 16 | 28.98663 |
| 406.12  | 5.3218165 | 33  | 21    | 12 | 32.68323 |
| 427.7   | 2.4124676 | 14  | 7     | 7  | 36.19020 |
| 450.3   | 4.6608954 | 23  | 17    | 6  | 36.19602 |
| 506.2   | 1.9330143 | 14  | 3     | 11 | 33.26623 |
| Canchan | 3.6665608 | 31  | 11    | 20 | 27.00126 |
| Desiree | 9.0626072 | 56  | 28    | 28 | 16.15569 |
| Unica   | 8.5447632 | 28  | 26    | 2  | 39.10400 |

*# With n = 4 and default ssi.method (farshadfar)*

**MASV.AMMI**(model, n = 4)

|        | MASV      | SSI | rMASV | rY | means    |
|--------|-----------|-----|-------|----|----------|
| 102.18 | 4.8247593 | 39  | 16    | 23 | 26.31947 |
| 104.22 | 4.0510711 | 23  | 10    | 13 | 31.28887 |
| 121.31 | 5.2473236 | 34  | 19    | 15 | 30.10174 |
| 141.28 | 5.9101338 | 23  | 22    | 1  | 39.75624 |
| 157.26 | 8.7719153 | 30  | 25    | 5  | 36.95181 |
| 163.9  | 4.5459209 | 41  | 14    | 27 | 21.41747 |
| 221.19 | 2.7137861 | 29  | 3     | 26 | 22.98480 |
| 233.11 | 3.7724279 | 26  | 9     | 17 | 28.66655 |
| 235.6  | 8.6953084 | 28  | 24    | 4  | 38.63477 |
| 241.2  | 2.8067193 | 26  | 4     | 22 | 26.34039 |
| 255.7  | 5.0424601 | 32  | 18    | 14 | 30.58975 |
| 314.12 | 3.4445298 | 25  | 7     | 18 | 28.17335 |
| 317.6  | 2.8792321 | 14  | 5     | 9  | 35.32583 |
| 319.20 | 8.8774217 | 30  | 27    | 3  | 38.75767 |
| 320.16 | 4.1787768 | 33  | 12    | 21 | 26.34808 |
| 342.15 | 4.1725070 | 35  | 11    | 24 | 26.01336 |
| 346.2  | 5.8554350 | 46  | 21    | 25 | 23.84175 |
| 351.26 | 6.4286626 | 31  | 23    | 8  | 36.11581 |
| 364.21 | 1.6075453 | 12  | 2     | 10 | 34.05974 |

```

402.7  0.5067415  20      1 19 27.47748
405.2  4.2896919  29     13 16 28.98663
406.12 5.3564283  32     20 12 32.68323
427.7  2.9737174  13      6  7 36.19020
450.3  4.7112537  21     15  6 36.19602
506.2  3.6306466  19      8 11 33.26623
Canchan 4.8979104  37     17 20 27.00126
Desiree 9.1023670  56     28 28 16.15569
Unica  8.7835476  28     26  2 39.10400

```

```

# With default n (N') and ssi.method = "rao"
MASV.AMMI(model, ssi.method = "rao")

```

|         | MASV      | SSI       | rMASV | rY | means    |
|---------|-----------|-----------|-------|----|----------|
| 102.18  | 4.7855876 | 1.4296717 | 19    | 23 | 26.31947 |
| 104.22  | 3.8328358 | 1.7337655 | 12    | 13 | 31.28887 |
| 121.31  | 4.0446758 | 1.6576851 | 14    | 15 | 30.10174 |
| 141.28  | 5.1867706 | 1.8235808 | 20    | 1  | 39.75624 |
| 157.26  | 7.6459224 | 1.5625443 | 24    | 5  | 36.95181 |
| 163.9   | 4.4977055 | 1.3064192 | 16    | 27 | 21.41747 |
| 221.19  | 2.1905344 | 1.9979910 | 5     | 26 | 22.98480 |
| 233.11  | 3.1794345 | 1.7949089 | 9     | 17 | 28.66655 |
| 235.6   | 8.4913020 | 1.5818054 | 25    | 4  | 38.63477 |
| 241.2   | 2.0338659 | 2.2035784 | 4     | 22 | 26.34039 |
| 255.7   | 4.7013868 | 1.5791422 | 18    | 14 | 30.58975 |
| 314.12  | 3.1376678 | 1.7902786 | 8     | 18 | 28.17335 |
| 317.6   | 2.3345492 | 2.3233562 | 6     | 9  | 35.32583 |
| 319.20  | 8.6398087 | 1.5802761 | 27    | 3  | 38.75767 |
| 320.16  | 3.8822326 | 1.5635888 | 13    | 21 | 26.34808 |
| 342.15  | 3.6438425 | 1.5987650 | 10    | 24 | 26.01336 |
| 346.2   | 5.3987165 | 1.2839782 | 22    | 25 | 23.84175 |
| 351.26  | 5.4005468 | 1.6840095 | 23    | 8  | 36.11581 |
| 364.21  | 1.4047546 | 3.0575043 | 2     | 10 | 34.05974 |
| 402.7   | 0.3537818 | 8.6266993 | 1     | 19 | 27.47748 |
| 405.2   | 4.1095727 | 1.6106479 | 15    | 16 | 28.98663 |
| 406.12  | 5.3218165 | 1.5795802 | 21    | 12 | 32.68323 |
| 427.7   | 2.4124676 | 2.3137009 | 7     | 7  | 36.19020 |
| 450.3   | 4.6608954 | 1.7669921 | 17    | 6  | 36.19602 |
| 506.2   | 1.9330143 | 2.4995588 | 3     | 11 | 33.26623 |
| Canchan | 3.6665608 | 1.6263253 | 11    | 20 | 27.00126 |
| Desiree | 9.0626072 | 0.8285565 | 28    | 28 | 16.15569 |
| Unica   | 8.5447632 | 1.5950896 | 26    | 2  | 39.10400 |

```

# Changing the ratio of weights for Rao's SSI
MASV.AMMI(model, ssi.method = "rao", a = 0.43)

```

|        | MASV      | SSI       | rMASV | rY | means    |
|--------|-----------|-----------|-------|----|----------|
| 102.18 | 4.7855876 | 1.1039112 | 19    | 23 | 26.31947 |
| 104.22 | 3.8328358 | 1.3270288 | 12    | 13 | 31.28887 |
| 121.31 | 4.0446758 | 1.2722512 | 14    | 15 | 30.10174 |
| 141.28 | 5.1867706 | 1.5230171 | 20    | 1  | 39.75624 |
| 157.26 | 7.6459224 | 1.3586506 | 24    | 5  | 36.95181 |
| 163.9  | 4.4977055 | 0.9598080 | 16    | 27 | 21.41747 |
| 221.19 | 2.1905344 | 1.2863130 | 5     | 26 | 22.98480 |
| 233.11 | 3.1794345 | 1.3045842 | 9     | 17 | 28.66655 |
| 235.6  | 8.4913020 | 1.3982110 | 25    | 4  | 38.63477 |

|         |           |           |    |    |          |
|---------|-----------|-----------|----|----|----------|
| 241.2   | 2.0338659 | 1.4370799 | 4  | 22 | 26.34039 |
| 255.7   | 4.7013868 | 1.2475474 | 18 | 14 | 30.58975 |
| 314.12  | 3.1376678 | 1.2934270 | 8  | 18 | 28.17335 |
| 317.6   | 2.3345492 | 1.6555805 | 6  | 9  | 35.32583 |
| 319.20  | 8.6398087 | 1.3998375 | 27 | 3  | 38.75767 |
| 320.16  | 3.8822326 | 1.1620273 | 13 | 21 | 26.34808 |
| 342.15  | 3.6438425 | 1.1709323 | 10 | 24 | 26.01336 |
| 346.2   | 5.3987165 | 0.9952142 | 22 | 25 | 23.84175 |
| 351.26  | 5.4005468 | 1.3953434 | 23 | 8  | 36.11581 |
| 364.21  | 1.4047546 | 1.9477337 | 2  | 10 | 34.05974 |
| 402.7   | 0.3537818 | 4.2201550 | 1  | 19 | 27.47748 |
| 405.2   | 4.1095727 | 1.2313006 | 15 | 16 | 28.98663 |
| 406.12  | 5.3218165 | 1.2866435 | 21 | 12 | 32.68323 |
| 427.7   | 2.4124676 | 1.6674932 | 7  | 7  | 36.19020 |
| 450.3   | 4.6608954 | 1.4325166 | 17 | 6  | 36.19602 |
| 506.2   | 1.9330143 | 1.6930696 | 3  | 11 | 33.26623 |
| Canchan | 3.6665608 | 1.2011435 | 11 | 20 | 27.00126 |
| Desiree | 9.0626072 | 0.6565359 | 28 | 28 | 16.15569 |
| Unica   | 8.5447632 | 1.4126439 | 26 | 2  | 39.10400 |

```
# With default n (N') and default ssi.method (farshadfar)
SIPC.AMMI(model)
```

```
SIPC.AMMI()
```

|         | SIPC      | SSI | rSIPC | rY | means    |
|---------|-----------|-----|-------|----|----------|
| 102.18  | 2.9592568 | 39  | 16    | 23 | 26.31947 |
| 104.22  | 2.2591593 | 22  | 9     | 13 | 31.28887 |
| 121.31  | 3.3872806 | 33  | 18    | 15 | 30.10174 |
| 141.28  | 4.3846248 | 23  | 22    | 1  | 39.75624 |
| 157.26  | 5.4846596 | 31  | 26    | 5  | 36.95181 |
| 163.9   | 2.6263670 | 38  | 11    | 27 | 21.41747 |
| 221.19  | 2.0218098 | 32  | 6     | 26 | 22.98480 |
| 233.11  | 2.1624442 | 24  | 7     | 17 | 28.66655 |
| 235.6   | 4.8273551 | 28  | 24    | 4  | 38.63477 |
| 241.2   | 2.0056410 | 27  | 5     | 22 | 26.34039 |
| 255.7   | 3.6075128 | 34  | 20    | 14 | 30.58975 |
| 314.12  | 2.4584089 | 28  | 10    | 18 | 28.17335 |
| 317.6   | 1.8698826 | 12  | 3     | 9  | 35.32583 |
| 319.20  | 5.9590451 | 31  | 28    | 3  | 38.75767 |
| 320.16  | 2.7040109 | 33  | 12    | 21 | 26.34808 |
| 342.15  | 2.9755899 | 41  | 17    | 24 | 26.01336 |
| 346.2   | 3.9525017 | 46  | 21    | 25 | 23.84175 |
| 351.26  | 4.5622439 | 31  | 23    | 8  | 36.11581 |
| 364.21  | 0.7526264 | 12  | 2     | 10 | 34.05974 |
| 402.7   | 0.2284995 | 20  | 1     | 19 | 27.47748 |
| 405.2   | 2.7952381 | 29  | 13    | 16 | 28.98663 |
| 406.12  | 2.8834753 | 27  | 15    | 12 | 32.68323 |
| 427.7   | 2.0049278 | 11  | 4     | 7  | 36.19020 |
| 450.3   | 2.8200387 | 20  | 14    | 6  | 36.19602 |
| 506.2   | 2.2178470 | 19  | 8     | 11 | 33.26623 |
| Canchan | 3.5328212 | 39  | 19    | 20 | 27.00126 |
| Desiree | 5.8073242 | 55  | 27    | 28 | 16.15569 |
| Unica   | 5.0654615 | 27  | 25    | 2  | 39.10400 |

```
# With n = 4 and default ssi.method (farshadfar)
SIPC.AMMI(model, n = 4)
```

|         | SIPC      | SSI | rSIPC | rY | means    |
|---------|-----------|-----|-------|----|----------|
| 102.18  | 3.4466455 | 38  | 15    | 23 | 26.31947 |
| 104.22  | 2.7007589 | 23  | 10    | 13 | 31.28887 |
| 121.31  | 5.6097497 | 38  | 23    | 15 | 30.10174 |
| 141.28  | 4.6372010 | 22  | 21    | 1  | 39.75624 |
| 157.26  | 7.4500476 | 33  | 28    | 5  | 36.95181 |
| 163.9   | 3.1338033 | 38  | 11    | 27 | 21.41747 |
| 221.19  | 2.1363292 | 29  | 3     | 26 | 22.98480 |
| 233.11  | 2.3911278 | 23  | 6     | 17 | 28.66655 |
| 235.6   | 5.8474857 | 29  | 25    | 4  | 38.63477 |
| 241.2   | 2.3056852 | 27  | 5     | 22 | 26.34039 |
| 255.7   | 3.9276052 | 31  | 17    | 14 | 30.58975 |
| 314.12  | 2.5182824 | 26  | 8     | 18 | 28.17335 |
| 317.6   | 2.4516869 | 16  | 7     | 9  | 35.32583 |
| 319.20  | 7.0781345 | 30  | 27    | 3  | 38.75767 |
| 320.16  | 4.0249810 | 39  | 18    | 21 | 26.34808 |
| 342.15  | 4.0957211 | 43  | 19    | 24 | 26.01336 |
| 346.2   | 4.8622465 | 47  | 22    | 25 | 23.84175 |
| 351.26  | 4.5974075 | 28  | 20    | 8  | 36.11581 |
| 364.21  | 1.5318314 | 12  | 2     | 10 | 34.05974 |
| 402.7   | 0.5893581 | 20  | 1     | 19 | 27.47748 |
| 405.2   | 3.3068718 | 29  | 13    | 16 | 28.98663 |
| 406.12  | 3.2694367 | 24  | 12    | 12 | 32.68323 |
| 427.7   | 2.5358269 | 16  | 9     | 7  | 36.19020 |
| 450.3   | 3.4327401 | 20  | 14    | 6  | 36.19602 |
| 506.2   | 2.2644412 | 15  | 4     | 11 | 33.26623 |
| Canchan | 3.6100050 | 36  | 16    | 20 | 27.00126 |
| Desiree | 5.8538044 | 54  | 26    | 28 | 16.15569 |
| Unica   | 5.7091275 | 26  | 24    | 2  | 39.10400 |

```
# With default n (N') and ssi.method = "rao"
SIPC.AMMI(model, ssi.method = "rao")
```

|        | SIPC      | SSI       | rSIPC | rY | means    |
|--------|-----------|-----------|-------|----|----------|
| 102.18 | 2.9592568 | 1.5124653 | 16    | 23 | 26.31947 |
| 104.22 | 2.2591593 | 1.8772594 | 9     | 13 | 31.28887 |
| 121.31 | 3.3872806 | 1.5531093 | 18    | 15 | 30.10174 |
| 141.28 | 4.3846248 | 1.7378762 | 22    | 1  | 39.75624 |
| 157.26 | 5.4846596 | 1.5578664 | 26    | 5  | 36.95181 |
| 163.9  | 2.6263670 | 1.4355650 | 11    | 27 | 21.41747 |
| 221.19 | 2.0218098 | 1.7071153 | 6     | 26 | 22.98480 |
| 233.11 | 2.1624442 | 1.8300896 | 7     | 17 | 28.66655 |
| 235.6  | 4.8273551 | 1.6608098 | 24    | 4  | 38.63477 |
| 241.2  | 2.0056410 | 1.8242469 | 5     | 22 | 26.34039 |
| 255.7  | 3.6075128 | 1.5341245 | 20    | 14 | 30.58975 |
| 314.12 | 2.4584089 | 1.7062126 | 10    | 18 | 28.17335 |
| 317.6  | 1.8698826 | 2.1873134 | 3     | 9  | 35.32583 |
| 319.20 | 5.9590451 | 1.5886436 | 28    | 3  | 38.75767 |
| 320.16 | 2.7040109 | 1.5751613 | 12    | 21 | 26.34808 |
| 342.15 | 2.9755899 | 1.4988930 | 17    | 24 | 26.01336 |
| 346.2  | 3.9525017 | 1.2672546 | 21    | 25 | 23.84175 |

```

351.26  4.5622439 1.6019853    23  8 36.11581
364.21  0.7526264 3.6831976     2 10 34.05974
402.7   0.2284995 9.3696848     1 19 27.47748
405.2   2.7952381 1.6378227    13 16 28.98663
406.12  2.8834753 1.7371554    15 12 32.68323
427.7   2.0049278 2.1457493     4  7 36.19020
450.3   2.8200387 1.8667975    14  6 36.19602
506.2   2.2178470 1.9576974     8 11 33.26623
Canchan 3.5328212 1.4284673    19 20 27.00126
Desiree 5.8073242 0.8601813    27 28 16.15569
Unica   5.0654615 1.6572552    25  2 39.10400

```

```

# Changing the ratio of weights for Rao's SSI
SIPC.AMMI(model, ssi.method = "rao", a = 0.43)

```

```

          SIPC      SSI rSIPC rY      means
102.18  2.9592568 1.1395125    16 23 26.31947
104.22  2.2591593 1.3887312     9 13 31.28887
121.31  3.3872806 1.2272836    18 15 30.10174
141.28  4.3846248 1.4861641    22  1 39.75624
157.26  5.4846596 1.3566391    26  5 36.95181
163.9   2.6263670 1.0153407    11 27 21.41747
221.19  2.0218098 1.1612364     6 26 22.98480
233.11  2.1624442 1.3197119     7 17 28.66655
235.6   4.8273551 1.4321829    24  4 38.63477
241.2   2.0056410 1.2739673     5 22 26.34039
255.7   3.6075128 1.2281898    20 14 30.58975
314.12  2.4584089 1.2572786    10 18 28.17335
317.6   1.8698826 1.5970821     3  9 35.32583
319.20  5.9590451 1.4034355    28  3 38.75767
320.16  2.7040109 1.1670035    12 21 26.34808
342.15  2.9755899 1.1279873    17 24 26.01336
346.2   3.9525017 0.9880230    21 25 23.84175
351.26  4.5622439 1.3600729    23  8 36.11581
364.21  0.7526264 2.2167818     2 10 34.05974
402.7   0.2284995 4.5396387     1 19 27.47748
405.2   2.7952381 1.2429858    13 16 28.98663
406.12  2.8834753 1.3544008    15 12 32.68323
427.7   2.0049278 1.5952740     4  7 36.19020
450.3   2.8200387 1.4754330    14  6 36.19602
506.2   2.2178470 1.4600692     8 11 33.26623
Canchan 3.5328212 1.1160645    19 20 27.00126
Desiree 5.8073242 0.6701345    27 28 16.15569
Unica   5.0654615 1.4393751    25  2 39.10400

```

```

# With default n (N') and default ssi.method (farshadfar)
ZA.AMMI(model)

```

```
ZA.AMMI()
```

```

          Za SSI rZa rY      means
102.18  0.15752787 41  18 23 26.31947
104.22  0.08552245 20   7 13 31.28887
121.31  0.13457796 26  11 15 30.10174
141.28  0.20424009 23  22  1 39.75624

```



```

157.26  0.20593889  28  23  5 36.95181
163.9   0.16161024  46  19  27 21.41747
221.19  0.08723440  34   8  26 22.98480
233.11  0.06559491  21   4  17 28.66655
235.6   0.20950908  29  25  4 38.63477
241.2   0.08160010  28   6  22 26.34039
255.7   0.16694984  34  20  14 30.58975
314.12  0.12243347  28  10  18 28.17335
317.6   0.08723605  18   9   9 35.32583
319.20  0.30778801  30  27   3 38.75767
320.16  0.14393358  35  14  21 26.34808
342.15  0.13891478  37  13  24 26.01336
346.2   0.20627243  49  24  25 23.84175
351.26  0.17809076  29  21   8 36.11581
364.21  0.03723882  12   2  10 34.05974
402.7   0.01243185  20   1  19 27.47748
405.2   0.15425031  33  17  16 28.98663
406.12  0.13595705  24  12  12 32.68323
427.7   0.07364374  12   5   7 36.19020
450.3   0.14895835  22  16   6 36.19602
506.2   0.06332050  14   3  11 33.26623
Canchan 0.14710608  35  15  20 27.00126
Desiree 0.32787182  56  28  28 16.15569
Unica   0.21646330  28  26   2 39.10400

```

```
# With n = 4 and default ssi.method (farshadfar)
```

```
ZA.AMMI(model, n = 4)
```

```

                Za SSI rZa rY    means
102.18  0.16239946  41  18  23 26.31947
104.22  0.08993636  21   8  13 31.28887
121.31  0.15679216  30  15  15 30.10174
141.28  0.20676466  23  22   1 39.75624
157.26  0.22558350  31  26   5 36.95181
163.9   0.16668221  46  19  27 21.41747
221.19  0.08837906  33   7  26 22.98480
233.11  0.06788066  21   4  17 28.66655
235.6   0.21970557  28  24   4 38.63477
241.2   0.08459913  28   6  22 26.34039
255.7   0.17014926  34  20  14 30.58975
314.12  0.12303192  28  10  18 28.17335
317.6   0.09305134  18   9   9 35.32583
319.20  0.31897363  30  27   3 38.75767
320.16  0.15713705  37  16  21 26.34808
342.15  0.15011080  37  13  24 26.01336
346.2   0.21536559  48  23  25 23.84175
351.26  0.17844223  29  21   8 36.11581
364.21  0.04502719  12   2  10 34.05974
402.7   0.01603874  20   1  19 27.47748
405.2   0.15936424  33  17  16 28.98663
406.12  0.13981485  23  11  12 32.68323
427.7   0.07895023  12   5   7 36.19020
450.3   0.15508247  20  14   6 36.19602
506.2   0.06378622  14   3  11 33.26623
Canchan 0.14787755  32  12  20 27.00126

```

```
Desiree 0.32833640 56 28 28 16.15569
Unica 0.22289692 27 25 2 39.10400
```

```
# With default n (N') and ssi.method = "rao"
ZA.AMMI(model, ssi.method = "rao")
```

|         | Za         | SSI       | rZa | rY | means    |
|---------|------------|-----------|-----|----|----------|
| 102.18  | 0.15752787 | 1.4309653 | 18  | 23 | 26.31947 |
| 104.22  | 0.08552245 | 2.0752658 | 7   | 13 | 31.28887 |
| 121.31  | 0.13457796 | 1.6519700 | 11  | 15 | 30.10174 |
| 141.28  | 0.20424009 | 1.7380721 | 22  | 1  | 39.75624 |
| 157.26  | 0.20593889 | 1.6429878 | 23  | 5  | 36.95181 |
| 163.9   | 0.16161024 | 1.2566633 | 19  | 27 | 21.41747 |
| 221.19  | 0.08723440 | 1.7838011 | 8   | 26 | 22.98480 |
| 233.11  | 0.06559491 | 2.3102920 | 4   | 17 | 28.66655 |
| 235.6   | 0.20950908 | 1.6903953 | 25  | 4  | 38.63477 |
| 241.2   | 0.08160010 | 1.9646329 | 6   | 22 | 26.34039 |
| 255.7   | 0.16694984 | 1.5378736 | 20  | 14 | 30.58975 |
| 314.12  | 0.12243347 | 1.6556010 | 10  | 18 | 28.17335 |
| 317.6   | 0.08723605 | 2.1861684 | 9   | 9  | 35.32583 |
| 319.20  | 0.30778801 | 1.5568815 | 27  | 3  | 38.75767 |
| 320.16  | 0.14393358 | 1.4859985 | 14  | 21 | 26.34808 |
| 342.15  | 0.13891478 | 1.4977340 | 13  | 24 | 26.01336 |
| 346.2   | 0.20627243 | 1.2148178 | 24  | 25 | 23.84175 |
| 351.26  | 0.17809076 | 1.6842433 | 21  | 8  | 36.11581 |
| 364.21  | 0.03723882 | 3.5336141 | 2   | 10 | 34.05974 |
| 402.7   | 0.01243185 | 8.1540882 | 1   | 19 | 27.47748 |
| 405.2   | 0.15425031 | 1.5301007 | 17  | 16 | 28.98663 |
| 406.12  | 0.13595705 | 1.7293399 | 12  | 12 | 32.68323 |
| 427.7   | 0.07364374 | 2.4052596 | 5   | 7  | 36.19020 |
| 450.3   | 0.14895835 | 1.7859494 | 16  | 6  | 36.19602 |
| 506.2   | 0.06332050 | 2.5096775 | 3   | 11 | 33.26623 |
| Canchan | 0.14710608 | 1.4937760 | 15  | 20 | 27.00126 |
| Desiree | 0.32787182 | 0.8019725 | 28  | 28 | 16.15569 |
| Unica   | 0.21646330 | 1.6918583 | 26  | 2  | 39.10400 |

```
# Changing the ratio of weights for Rao's SSI
ZA.AMMI(model, ssi.method = "rao", a = 0.43)
```

|        | Za         | SSI       | rZa | rY | means    |
|--------|------------|-----------|-----|----|----------|
| 102.18 | 0.15752787 | 1.1044675 | 18  | 23 | 26.31947 |
| 104.22 | 0.08552245 | 1.4738739 | 7   | 13 | 31.28887 |
| 121.31 | 0.13457796 | 1.2697937 | 11  | 15 | 30.10174 |
| 141.28 | 0.20424009 | 1.4862483 | 22  | 1  | 39.75624 |
| 157.26 | 0.20593889 | 1.3932413 | 23  | 5  | 36.95181 |
| 163.9  | 0.16161024 | 0.9384129 | 19  | 27 | 21.41747 |
| 221.19 | 0.08723440 | 1.1942113 | 8   | 26 | 22.98480 |
| 233.11 | 0.06559491 | 1.5261989 | 4   | 17 | 28.66655 |
| 235.6  | 0.20950908 | 1.4449047 | 25  | 4  | 38.63477 |
| 241.2  | 0.08160010 | 1.3343333 | 6   | 22 | 26.34039 |
| 255.7  | 0.16694984 | 1.2298019 | 20  | 14 | 30.58975 |
| 314.12 | 0.12243347 | 1.2355156 | 10  | 18 | 28.17335 |
| 317.6  | 0.08723605 | 1.5965898 | 9   | 9  | 35.32583 |
| 319.20 | 0.30778801 | 1.3897778 | 27  | 3  | 38.75767 |
| 320.16 | 0.14393358 | 1.1286635 | 14  | 21 | 26.34808 |
| 342.15 | 0.13891478 | 1.1274889 | 13  | 24 | 26.01336 |

|         |            |           |    |    |          |
|---------|------------|-----------|----|----|----------|
| 346.2   | 0.20627243 | 0.9654752 | 24 | 25 | 23.84175 |
| 351.26  | 0.17809076 | 1.3954439 | 21 | 8  | 36.11581 |
| 364.21  | 0.03723882 | 2.1524610 | 2  | 10 | 34.05974 |
| 402.7   | 0.01243185 | 4.0169322 | 1  | 19 | 27.47748 |
| 405.2   | 0.15425031 | 1.1966653 | 17 | 16 | 28.98663 |
| 406.12  | 0.13595705 | 1.3510402 | 12 | 12 | 32.68323 |
| 427.7   | 0.07364374 | 1.7068634 | 5  | 7  | 36.19020 |
| 450.3   | 0.14895835 | 1.4406683 | 16 | 6  | 36.19602 |
| 506.2   | 0.06332050 | 1.6974207 | 3  | 11 | 33.26623 |
| Canchan | 0.14710608 | 1.1441472 | 15 | 20 | 27.00126 |
| Desiree | 0.32787182 | 0.6451047 | 28 | 28 | 16.15569 |
| Unica   | 0.21646330 | 1.4542544 | 26 | 2  | 39.10400 |

### Simultaneous selection indices for yield and stability

The most stable genotype need not necessarily be the highest yielding genotype. Hence, simultaneous selection indices (SSIs) have been proposed for the selection of stable as well as high yielding genotypes.

A family of simultaneous selection indices ( $I_i$ ) were proposed by Rao and Prabhakaran (2005) similar to those proposed by Bajpai and Prabhakaran (2000) by incorporating the AMMI Based Stability Parameter ( $ASTAB$ ) and Yield as components. These indices consist of yield component, measured as the ratio of the average performance of the  $i$ th genotype to the overall mean performance of the genotypes under test and a stability component, measured as the ratio of stability information ( $\frac{1}{ASTAB}$ ) of the  $i$ th genotype to the mean stability information of the genotypes under test.

$$I_i = \frac{\bar{Y}_i}{\bar{Y}_{..}} + \alpha \frac{\frac{1}{ASTAB_i}}{\frac{1}{T} \sum_{i=1}^T \frac{1}{ASTAB_i}}$$

Where  $ASTAB_i$  is the stability measure of the  $i$ th genotype under AMMI procedure;  $Y_i$  is mean performance of  $i$ th genotype;  $Y_{..}$  is the overall mean;  $T$  is the number of genotypes under test and  $\alpha$  is the ratio of the weights given to the stability components ( $w_2$ ) and yield ( $w_1$ ) with a restriction that  $w_1 + w_2 = 1$ . The weights can be specified as required (Table 2).

**Table 2 :**  $\alpha$  and corresponding weights ( $w_1$  and  $w_2$ )

| $\alpha$ | $w_1$ | $w_2$ |
|----------|-------|-------|
| 1.00     | 0.5   | 0.5   |
| 0.67     | 0.6   | 0.4   |
| 0.43     | 0.7   | 0.3   |
| 0.25     | 0.8   | 0.2   |

In `ammistability`, the above expression has been implemented for all the stability parameters ( $SP$ ) including  $ASTAB$ .

$$I_i = \frac{\bar{Y}_i}{\bar{Y}_{..}} + \alpha \frac{\frac{1}{SP_i}}{\frac{1}{T} \sum_{i=1}^T \frac{1}{SP_i}}$$

Genotype stability index ( $GSI$ ) (Farshadfar, 2008) or Yield stability index ( $YSI$ ) (Farshadfar et al., 2011; Jambhulkar et al., 2017) is a simultaneous selection index for yield and yield stability which is computed by summation of the ranks of the stability index/parameter and the ranks of the mean yields.  $YSI$  is computed for all the stability parameters/indices implemented in this package.

$$GSI = YSI = R_{SP} + R_Y$$

Where,  $R_{SP}$  is the stability parameter/index rank of the genotype and  $R_Y$  is the mean yield rank of the genotype.

The function SSI implements both these indices in `ammistability`. Further, for each of the stability parameter functions, the simultaneous selection index is also computed by either of these functions as specified by the argument `ssi.method`.

## Examples

```
library(agricolae)
data(plrv)
model <- with(plrv, AMMI(Localty, Genotype, Rep, Yield, console=FALSE))

yield <- aggregate(model$means$Yield, by= list(model$means$GEN),
                    FUN=mean, na.rm=TRUE)[,2]
stab <- DZ.AMMI(model)$DZ
genotypes <- rownames(DZ.AMMI(model))

# With default ssi.method (farshadfar)
SSI(y = yield, sp = stab, gen = genotypes)
```

SSI()

|         | SP         | SSI | rSP | rY | means    |
|---------|------------|-----|-----|----|----------|
| 102.18  | 0.26393535 | 37  | 14  | 23 | 26.31947 |
| 104.22  | 0.22971564 | 21  | 8   | 13 | 31.28887 |
| 121.31  | 0.32031744 | 34  | 19  | 15 | 30.10174 |
| 141.28  | 0.39838535 | 23  | 22  | 1  | 39.75624 |
| 157.26  | 0.53822924 | 33  | 28  | 5  | 36.95181 |
| 163.9   | 0.26659011 | 42  | 15  | 27 | 21.41747 |
| 221.19  | 0.19563325 | 29  | 3   | 26 | 22.98480 |
| 233.11  | 0.25167755 | 27  | 10  | 17 | 28.66655 |
| 235.6   | 0.46581370 | 28  | 24  | 4  | 38.63477 |
| 241.2   | 0.21481887 | 28  | 6   | 22 | 26.34039 |
| 255.7   | 0.30862904 | 31  | 17  | 14 | 30.58975 |
| 314.12  | 0.22603261 | 25  | 7   | 18 | 28.17335 |
| 317.6   | 0.20224771 | 14  | 5   | 9  | 35.32583 |
| 319.20  | 0.50675112 | 29  | 26  | 3  | 38.75767 |
| 320.16  | 0.23280596 | 30  | 9   | 21 | 26.34808 |
| 342.15  | 0.25989774 | 36  | 12  | 24 | 26.01336 |
| 346.2   | 0.37125512 | 45  | 20  | 25 | 23.84175 |
| 351.26  | 0.43805896 | 31  | 23  | 8  | 36.11581 |
| 364.21  | 0.07409309 | 12  | 2   | 10 | 34.05974 |
| 402.7   | 0.02004533 | 20  | 1   | 19 | 27.47748 |
| 405.2   | 0.26238837 | 29  | 13  | 16 | 28.98663 |
| 406.12  | 0.28179394 | 28  | 16  | 12 | 32.68323 |
| 427.7   | 0.20176581 | 11  | 4   | 7  | 36.19020 |
| 450.3   | 0.25465368 | 17  | 11  | 6  | 36.19602 |
| 506.2   | 0.30899851 | 29  | 18  | 11 | 33.26623 |
| Canchan | 0.37201039 | 41  | 21  | 20 | 27.00126 |
| Desiree | 0.52005815 | 55  | 27  | 28 | 16.15569 |

```
Unica 0.48083049 27 25 2 39.10400
```

```
# With ssi.method = "rao"
```

```
SSI(y = yield, sp = stab, gen = genotypes, method = "rao")
```

|         | SP         | SSI        | rSP | rY | means    |
|---------|------------|------------|-----|----|----------|
| 102.18  | 0.26393535 | 1.5536988  | 14  | 23 | 26.31947 |
| 104.22  | 0.22971564 | 1.8193399  | 8   | 13 | 31.28887 |
| 121.31  | 0.32031744 | 1.5545939  | 19  | 15 | 30.10174 |
| 141.28  | 0.39838535 | 1.7570779  | 22  | 1  | 39.75624 |
| 157.26  | 0.53822924 | 1.5459114  | 28  | 5  | 36.95181 |
| 163.9   | 0.26659011 | 1.3869397  | 15  | 27 | 21.41747 |
| 221.19  | 0.19563325 | 1.6878048  | 3   | 26 | 22.98480 |
| 233.11  | 0.25167755 | 1.6641025  | 10  | 17 | 28.66655 |
| 235.6   | 0.46581370 | 1.6538090  | 24  | 4  | 38.63477 |
| 241.2   | 0.21481887 | 1.7134093  | 6   | 22 | 26.34039 |
| 255.7   | 0.30862904 | 1.5922105  | 17  | 14 | 30.58975 |
| 314.12  | 0.22603261 | 1.7307783  | 7   | 18 | 28.17335 |
| 317.6   | 0.20224771 | 2.0595024  | 5   | 9  | 35.32583 |
| 319.20  | 0.50675112 | 1.6259792  | 26  | 3  | 38.75767 |
| 320.16  | 0.23280596 | 1.6476346  | 9   | 21 | 26.34808 |
| 342.15  | 0.25989774 | 1.5545233  | 12  | 24 | 26.01336 |
| 346.2   | 0.37125512 | 1.2718506  | 20  | 25 | 23.84175 |
| 351.26  | 0.43805896 | 1.5966462  | 23  | 8  | 36.11581 |
| 364.21  | 0.07409309 | 3.5881882  | 2   | 10 | 34.05974 |
| 402.7   | 0.02004533 | 10.0539968 | 1   | 19 | 27.47748 |
| 405.2   | 0.26238837 | 1.6447637  | 13  | 16 | 28.98663 |
| 406.12  | 0.28179394 | 1.7171135  | 16  | 12 | 32.68323 |
| 427.7   | 0.20176581 | 2.0898536  | 4   | 7  | 36.19020 |
| 450.3   | 0.25465368 | 1.9010808  | 11  | 6  | 36.19602 |
| 506.2   | 0.30899851 | 1.6787677  | 18  | 11 | 33.26623 |
| Canchan | 0.37201039 | 1.3738642  | 21  | 20 | 27.00126 |
| Desiree | 0.52005815 | 0.8797586  | 27  | 28 | 16.15569 |
| Unica   | 0.48083049 | 1.6568004  | 25  | 2  | 39.10400 |

```
# Changing the ratio of weights for Rao's SSI
```

```
SSI(y = yield, sp = stab, gen = genotypes, method = "rao", a = 0.43)
```

|        | SP         | SSI       | rSP | rY | means    |
|--------|------------|-----------|-----|----|----------|
| 102.18 | 0.26393535 | 1.1572429 | 14  | 23 | 26.31947 |
| 104.22 | 0.22971564 | 1.3638258 | 8   | 13 | 31.28887 |
| 121.31 | 0.32031744 | 1.2279220 | 19  | 15 | 30.10174 |
| 141.28 | 0.39838535 | 1.4944208 | 22  | 1  | 39.75624 |
| 157.26 | 0.53822924 | 1.3514985 | 28  | 5  | 36.95181 |
| 163.9  | 0.26659011 | 0.9944318 | 15  | 27 | 21.41747 |
| 221.19 | 0.19563325 | 1.1529329 | 3   | 26 | 22.98480 |
| 233.11 | 0.25167755 | 1.2483375 | 10  | 17 | 28.66655 |
| 235.6  | 0.46581370 | 1.4291726 | 24  | 4  | 38.63477 |
| 241.2  | 0.21481887 | 1.2263072 | 6   | 22 | 26.34039 |
| 255.7  | 0.30862904 | 1.2531668 | 17  | 14 | 30.58975 |
| 314.12 | 0.22603261 | 1.2678419 | 7   | 18 | 28.17335 |
| 317.6  | 0.20224771 | 1.5421234 | 5   | 9  | 35.32583 |
| 319.20 | 0.50675112 | 1.4194898 | 26  | 3  | 38.75767 |
| 320.16 | 0.23280596 | 1.1981670 | 9   | 21 | 26.34808 |
| 342.15 | 0.25989774 | 1.1519083 | 12  | 24 | 26.01336 |

|         |            |           |    |    |          |
|---------|------------|-----------|----|----|----------|
| 346.2   | 0.37125512 | 0.9899993 | 20 | 25 | 23.84175 |
| 351.26  | 0.43805896 | 1.3577771 | 23 | 8  | 36.11581 |
| 364.21  | 0.07409309 | 2.1759278 | 2  | 10 | 34.05974 |
| 402.7   | 0.02004533 | 4.8338929 | 1  | 19 | 27.47748 |
| 405.2   | 0.26238837 | 1.2459704 | 13 | 16 | 28.98663 |
| 406.12  | 0.28179394 | 1.3457828 | 16 | 12 | 32.68323 |
| 427.7   | 0.20176581 | 1.5712389 | 4  | 7  | 36.19020 |
| 450.3   | 0.25465368 | 1.4901748 | 11 | 6  | 36.19602 |
| 506.2   | 0.30899851 | 1.3401295 | 18 | 11 | 33.26623 |
| Canchan | 0.37201039 | 1.0925852 | 21 | 20 | 27.00126 |
| Desiree | 0.52005815 | 0.6785528 | 27 | 28 | 16.15569 |
| Unica   | 0.48083049 | 1.4391795 | 25 | 2  | 39.10400 |

## Wrapper function

A function `ammistability` has also been implemented which is a wrapper around all the available functions in the package to compute simultaneously multiple AMMI stability parameters along with the corresponding SSIs. Correlation among the computed values as well as visualization of the differences in genotype ranks for the computed parameters is also generated.

## Examples

```
library(agricolae)
data(plrv)

# AMMI model
model <- with(plrv, AMMI(Locality, Genotype, Rep, Yield, console = FALSE))

ammistability(model, AMGE = TRUE, ASI = FALSE, ASV = TRUE, ASTAB = FALSE,
               AVAMGE = FALSE, DA = FALSE, DZ = FALSE, EV = TRUE,
               FA = FALSE, MASI = FALSE, MASV = TRUE, SIPC = TRUE,
               ZA = FALSE)
```

```
ammistability()
```

```
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
i Please use tidy evaluation idioms with `aes()`.
i See also `vignette("ggplot2-in-packages")` for more information.
i The deprecated feature was likely used in the ggcorrplot package.
  Please report the issue at <https://github.com/kassambara/ggcorrplot/issues>.
This warning is displayed once every 8 hours.
Call `lifecycle::last_lifecycle_warnings()` to see where this warning was generated.

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
i The deprecated feature was likely used in the ammistability package.
  Please report the issue at <https://github.com/ajaygpb/ammistability/issues>.
This warning is displayed once every 8 hours.
Call `lifecycle::last_lifecycle_warnings()` to see where this warning was generated.
```

```
$Details
$Details$`Stability parameters estimated`
[1] "AMGE" "ASV"  "EV"   "MASV" "SIPC"

$Details$`SSI method`
```

[1] "Farshadfar (2008)"

# \$`Stability Parameters`

|    | genotype | means    | AMGE          | ASV       | EV           | MASV      | SIPC      |
|----|----------|----------|---------------|-----------|--------------|-----------|-----------|
| 1  | 102.18   | 26.31947 | 6.661338e-15  | 3.3801820 | 0.0232206231 | 4.7855876 | 2.9592568 |
| 2  | 104.22   | 31.28887 | 1.110223e-15  | 1.4627695 | 0.0175897578 | 3.8328358 | 2.2591593 |
| 3  | 121.31   | 30.10174 | 1.199041e-14  | 2.2937918 | 0.0342010876 | 4.0446758 | 3.3872806 |
| 4  | 141.28   | 39.75624 | -1.110223e-14 | 4.4672401 | 0.0529036285 | 5.1867706 | 4.3846248 |
| 5  | 157.26   | 36.95181 | 2.131628e-14  | 3.2923168 | 0.0965635719 | 7.6459224 | 5.4846596 |
| 6  | 163.9    | 21.41747 | 5.329071e-15  | 4.4269636 | 0.0236900961 | 4.4977055 | 2.6263670 |
| 7  | 221.19   | 22.98480 | 9.325873e-15  | 1.8014494 | 0.0127574566 | 2.1905344 | 2.0218098 |
| 8  | 233.11   | 28.66655 | -1.088019e-14 | 1.0582263 | 0.0211138628 | 3.1794345 | 2.1624442 |
| 9  | 235.6    | 38.63477 | -1.687539e-14 | 3.7647078 | 0.0723274691 | 8.4913020 | 4.8273551 |
| 10 | 241.2    | 26.34039 | 9.325873e-15  | 1.6774241 | 0.0153823821 | 2.0338659 | 2.0056410 |
| 11 | 255.7    | 30.58975 | -8.215650e-15 | 3.3289736 | 0.0317506280 | 4.7013868 | 3.6075128 |
| 12 | 314.12   | 28.17335 | 0.000000e+00  | 2.9170536 | 0.0170302467 | 3.1376678 | 2.4584089 |
| 13 | 317.6    | 35.32583 | 4.884981e-15  | 2.1874274 | 0.0136347120 | 2.3345492 | 1.8698826 |
| 14 | 319.20   | 38.75767 | -7.105427e-15 | 6.7164864 | 0.0855988994 | 8.6398087 | 5.9590451 |
| 15 | 320.16   | 26.34808 | 7.993606e-15  | 3.3208950 | 0.0180662044 | 3.8822326 | 2.7040109 |
| 16 | 342.15   | 26.01336 | -7.327472e-15 | 2.9219360 | 0.0225156118 | 3.6438425 | 2.9755899 |
| 17 | 346.2    | 23.84175 | -6.217249e-15 | 5.1827747 | 0.0459434537 | 5.3987165 | 3.9525017 |
| 18 | 351.26   | 36.11581 | -8.659740e-15 | 2.9786832 | 0.0639652186 | 5.4005468 | 4.5622439 |
| 19 | 364.21   | 34.05974 | -1.915135e-15 | 0.7236998 | 0.0018299284 | 1.4047546 | 0.7526264 |
| 20 | 402.7    | 27.47748 | -2.636780e-16 | 0.2801470 | 0.0001339385 | 0.3537818 | 0.2284995 |
| 21 | 405.2    | 28.98663 | 4.440892e-16  | 3.9832546 | 0.0229492190 | 4.1095727 | 2.7952381 |
| 22 | 406.12   | 32.68323 | -5.995204e-15 | 2.5631734 | 0.0264692745 | 5.3218165 | 2.8834753 |
| 23 | 427.7    | 36.19020 | 3.219647e-15  | 1.1467970 | 0.0135698145 | 2.4124676 | 2.0049278 |
| 24 | 450.3    | 36.19602 | -4.440892e-16 | 3.1430174 | 0.0216161656 | 4.6608954 | 2.8200387 |
| 25 | 506.2    | 33.26623 | 1.176836e-14  | 0.7511331 | 0.0318266934 | 1.9330143 | 2.2178470 |
| 26 | Canchan  | 27.00126 | -1.421085e-14 | 3.0975884 | 0.0461305761 | 3.6665608 | 3.5328212 |
| 27 | Desiree  | 16.15569 | 1.088019e-14  | 7.7833445 | 0.0901534938 | 9.0626072 | 5.8073242 |
| 28 | Unica    | 39.10400 | -2.664535e-15 | 3.8380782 | 0.0770659860 | 8.5447632 | 5.0654615 |

# \$`Simultaneous Selection Indices`

|    | genotype | means    | AMGE_SSI | ASV_SSI | EV_SSI | MASV_SSI | SIPC_SSI |
|----|----------|----------|----------|---------|--------|----------|----------|
| 1  | 102.18   | 26.31947 | 44.0     | 43      | 37     | 42       | 39       |
| 2  | 104.22   | 31.28887 | 30.0     | 19      | 21     | 25       | 22       |
| 3  | 121.31   | 30.10174 | 42.0     | 25      | 34     | 29       | 33       |
| 4  | 141.28   | 39.75624 | 4.0      | 26      | 23     | 21       | 23       |
| 5  | 157.26   | 36.95181 | 33.0     | 22      | 33     | 29       | 31       |
| 6  | 163.9    | 21.41747 | 47.0     | 51      | 42     | 43       | 38       |
| 7  | 221.19   | 22.98480 | 49.5     | 34      | 29     | 31       | 32       |
| 8  | 233.11   | 28.66655 | 21.0     | 21      | 27     | 26       | 24       |
| 9  | 235.6    | 38.63477 | 5.0      | 25      | 28     | 29       | 28       |
| 10 | 241.2    | 26.34039 | 45.5     | 29      | 28     | 26       | 27       |
| 11 | 255.7    | 30.58975 | 20.0     | 33      | 31     | 32       | 34       |
| 12 | 314.12   | 28.17335 | 33.0     | 30      | 25     | 26       | 28       |
| 13 | 317.6    | 35.32583 | 28.0     | 18      | 14     | 15       | 12       |
| 14 | 319.20   | 38.75767 | 11.0     | 30      | 29     | 30       | 31       |
| 15 | 320.16   | 26.34808 | 43.0     | 39      | 30     | 34       | 33       |
| 16 | 342.15   | 26.01336 | 31.0     | 37      | 36     | 34       | 41       |
| 17 | 346.2    | 23.84175 | 34.0     | 51      | 45     | 47       | 46       |
| 18 | 351.26   | 36.11581 | 13.0     | 22      | 31     | 31       | 31       |

|    |         |          |      |    |    |    |    |
|----|---------|----------|------|----|----|----|----|
| 19 | 364.21  | 34.05974 | 22.0 | 12 | 12 | 12 | 12 |
| 20 | 402.7   | 27.47748 | 33.0 | 20 | 20 | 20 | 20 |
| 21 | 405.2   | 28.98663 | 32.0 | 39 | 29 | 31 | 29 |
| 22 | 406.12  | 32.68323 | 22.0 | 23 | 28 | 33 | 27 |
| 23 | 427.7   | 36.19020 | 25.0 | 12 | 11 | 14 | 11 |
| 24 | 450.3   | 36.19602 | 19.0 | 22 | 17 | 23 | 20 |
| 25 | 506.2   | 33.26623 | 37.0 | 14 | 29 | 14 | 19 |
| 26 | Canchan | 27.00126 | 22.0 | 35 | 41 | 31 | 39 |
| 27 | Desiree | 16.15569 | 53.0 | 56 | 55 | 56 | 55 |
| 28 | Unica   | 39.10400 | 13.0 | 24 | 27 | 28 | 27 |

## \$`SP Correlation`

|      | AMGE   | ASV    | EV     | MASV   | SIPC   |
|------|--------|--------|--------|--------|--------|
| AMGE | 1.00** | <NA>   | <NA>   | <NA>   | <NA>   |
| ASV  | -0.10  | 1.00** | <NA>   | <NA>   | <NA>   |
| EV   | -0.06  | 0.70** | 1.00** | <NA>   | <NA>   |
| MASV | -0.14  | 0.81** | 0.90** | 1.00** | <NA>   |
| SIPC | -0.12  | 0.81** | 0.96** | 0.94** | 1.00** |

## \$`SSI Correlation`

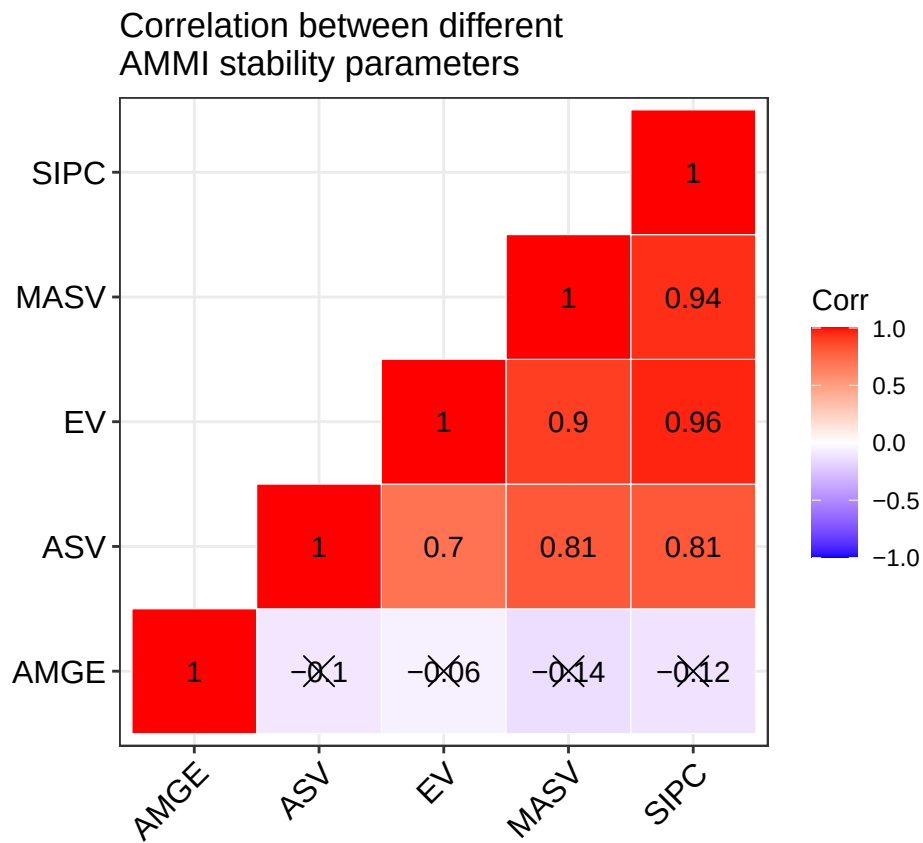
|      | AMGE   | ASV    | EV     | MASV   | SIPC   |
|------|--------|--------|--------|--------|--------|
| AMGE | 1.00** | <NA>   | <NA>   | <NA>   | <NA>   |
| ASV  | 0.49** | 1.00** | <NA>   | <NA>   | <NA>   |
| EV   | 0.43*  | 0.84** | 1.00** | <NA>   | <NA>   |
| MASV | 0.41*  | 0.92** | 0.90** | 1.00** | <NA>   |
| SIPC | 0.40*  | 0.89** | 0.96** | 0.95** | 1.00** |

## \$`SP and SSI Correlation`

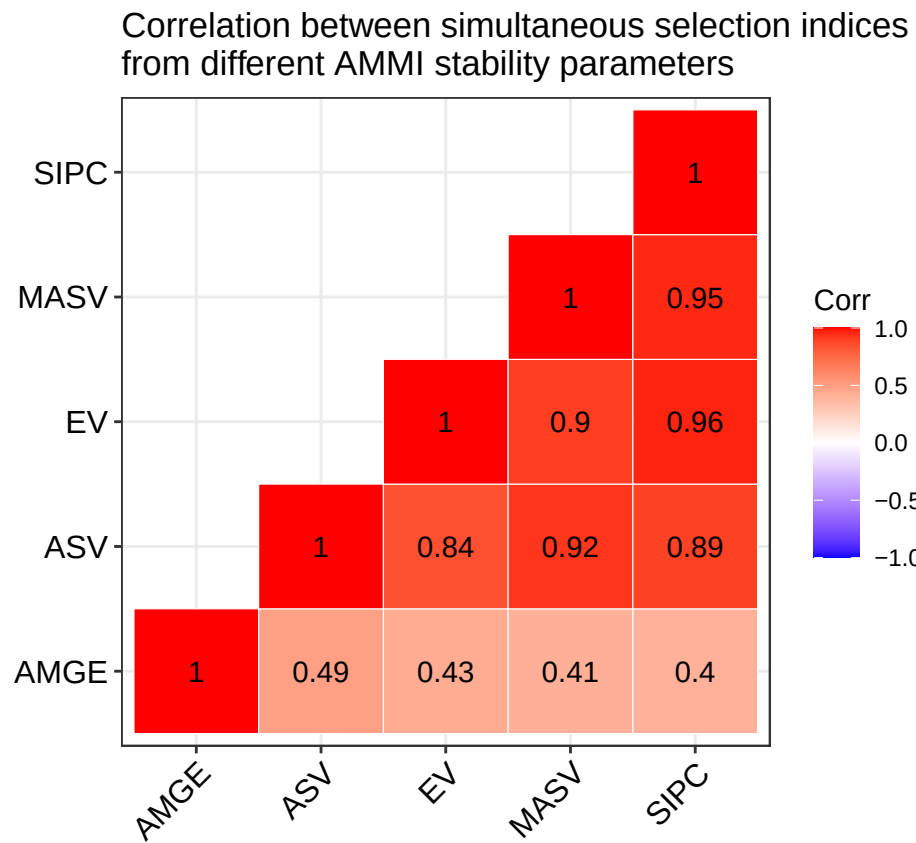
|          | AMGE   | ASV    | EV     | MASV   | SIPC   | AMGE_SSI | ASV_SSI | EV_SSI | MASV_SSI | SIPC_SSI |
|----------|--------|--------|--------|--------|--------|----------|---------|--------|----------|----------|
| AMGE     | 1.00** | <NA>   | <NA>   | <NA>   | <NA>   | <NA>     | <NA>    | <NA>   | <NA>     | <NA>     |
| ASV      | -0.10  | 1.00** | <NA>   | <NA>   | <NA>   | <NA>     | <NA>    | <NA>   | <NA>     | <NA>     |
| EV       | -0.06  | 0.70** | 1.00** | <NA>   | <NA>   | <NA>     | <NA>    | <NA>   | <NA>     | <NA>     |
| MASV     | -0.14  | 0.81** | 0.90** | 1.00** | <NA>   | <NA>     | <NA>    | <NA>   | <NA>     | <NA>     |
| SIPC     | -0.12  | 0.81** | 0.96** | 0.94** | 1.00** | <NA>     | <NA>    | <NA>   | <NA>     | <NA>     |
| AMGE_SSI | 0.76** | -0.03  | -0.30  | -0.30  | -0.29  | 1.00**   | <NA>    | <NA>   | <NA>     | <NA>     |
| ASV_SSI  | 0.05   | 0.71** | 0.21   | 0.35   | 0.34   | 0.49**   | 1.00**  | <NA>   | <NA>     | <NA>     |
| EV_SSI   | 0.09   | 0.64** | 0.48** | 0.47*  | 0.53** | 0.43*    | 0.84**  | 1.00** | <NA>     | <NA>     |
| MASV_SSI | 0.04   | 0.73** | 0.40*  | 0.54** | 0.51** | 0.41*    | 0.92**  | 0.90** | 1.00**   | <NA>     |
| SIPC_SSI | 0.02   | 0.70** | 0.45*  | 0.50** | 0.54** | 0.40*    | 0.89**  | 0.96** | 0.95**   | 1.00**   |

## \$`SP Correlogram`

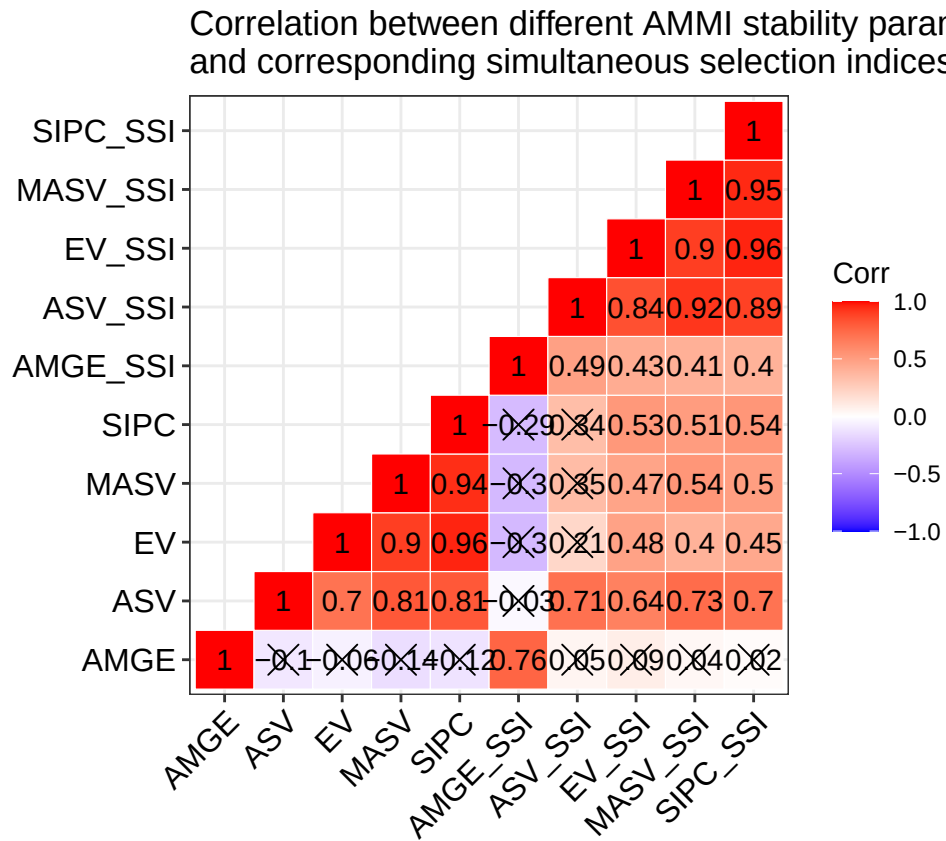




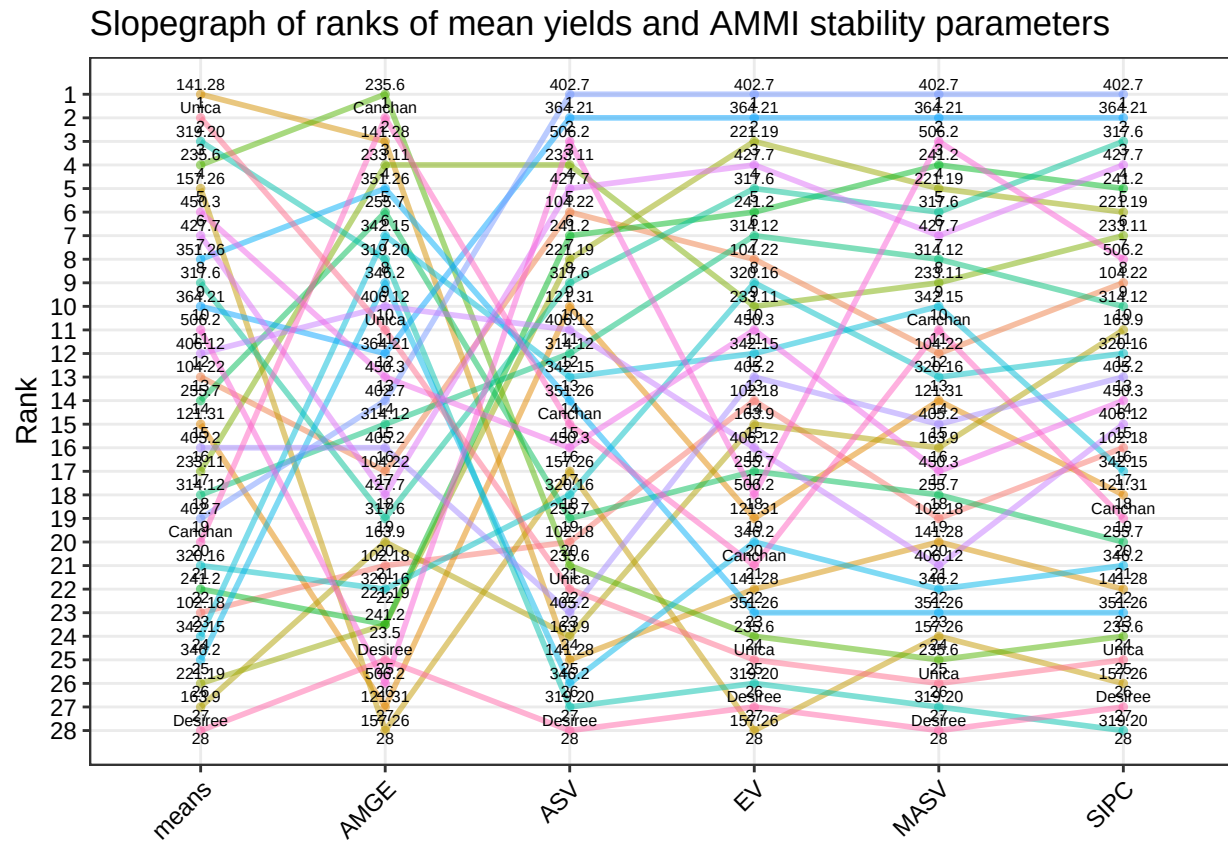
\$`SSI Correlogram`



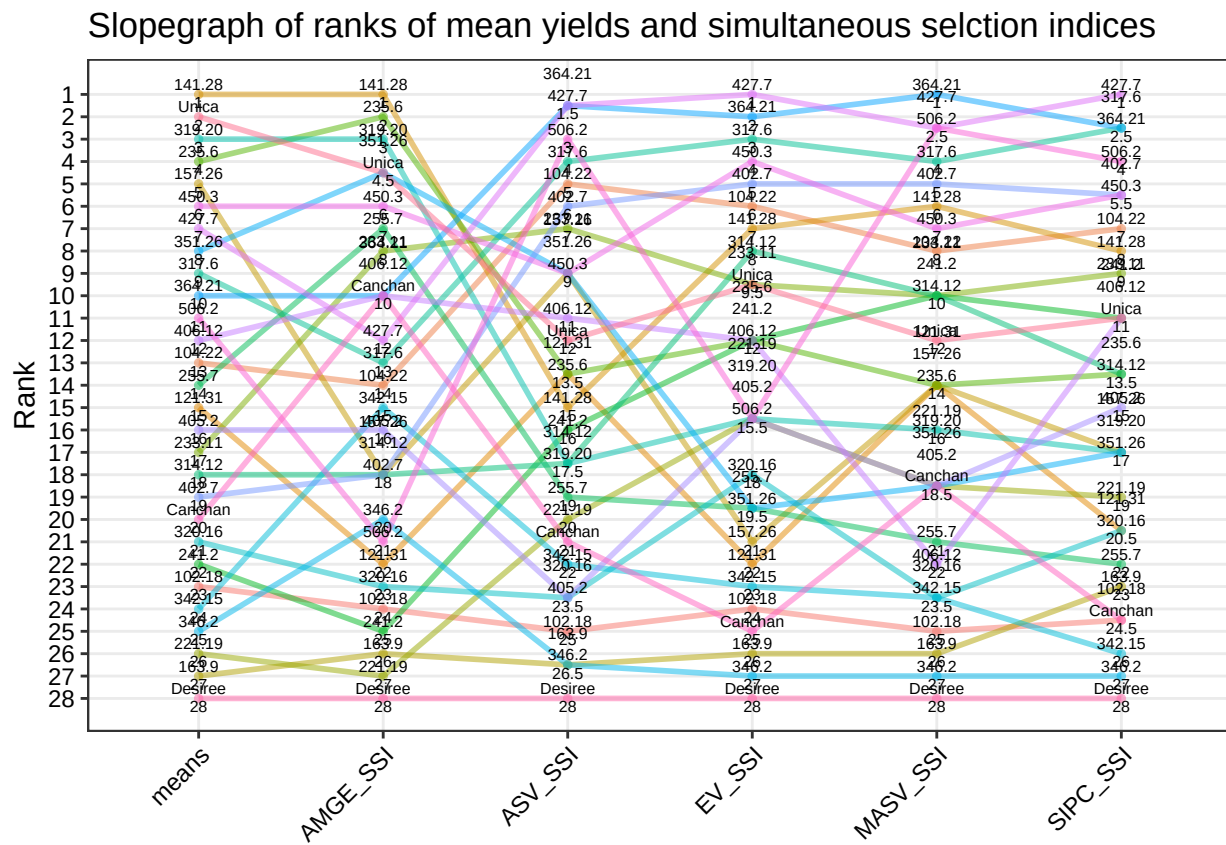
\$`SP and SSI Correlogram`



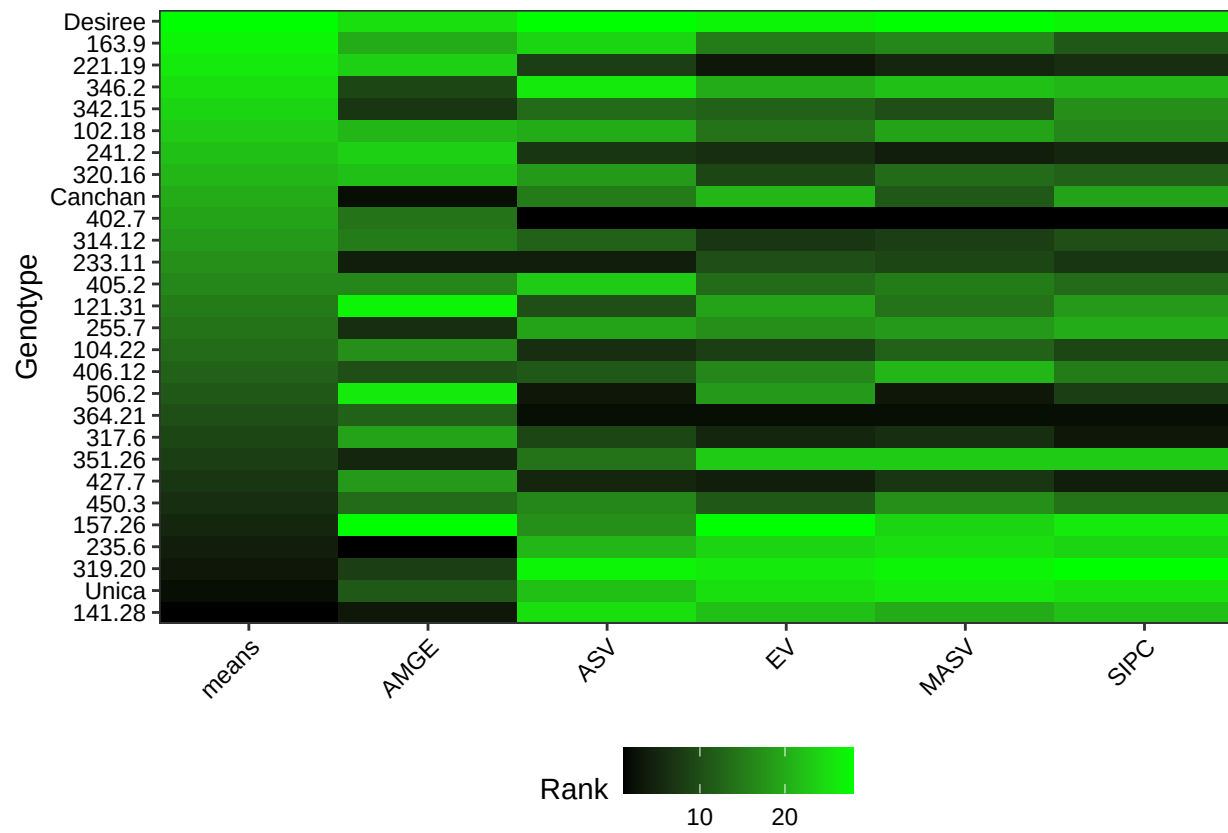
\$`SP Slopegraph`



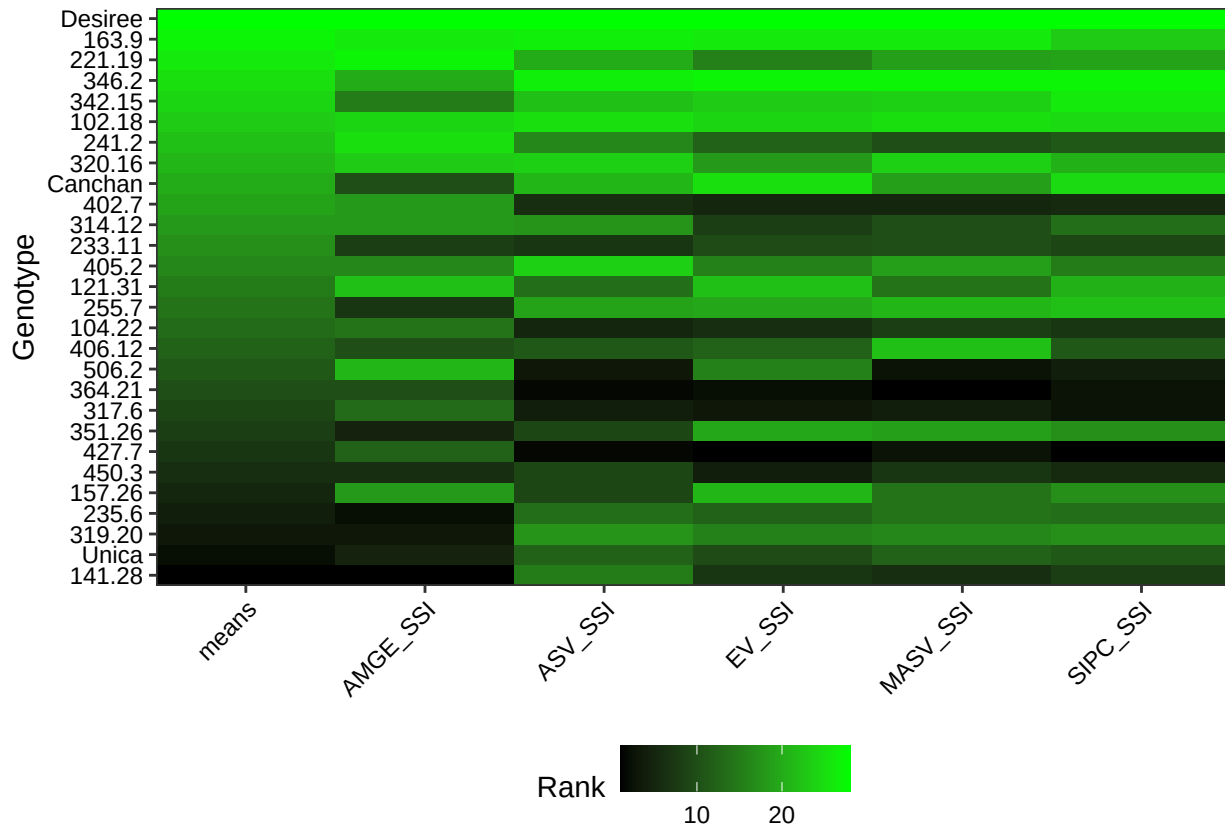
\$`SSI Slopegraph`



\$`SP Heatmap`



\$`SSI Heatmap`



## Citing *ammistability*

To cite the R package '*ammistability*' in publications use:

Ajay, B. C., Aravind, J., and Abdul Fiyaz, R. (2019). *ammistability*: R package for ranking genotypes based on stability parameters derived from AMMI model. *Indian Journal of Genetics and Plant Breeding (The)*, 79(2), 460-466.  
<https://www.isgpb.org/article/ammistability-r-package-for-ranking-genotypes-based-on-stability-paramete>

Ajay, B. C., Aravind, J., and Abdul Fiyaz, R. (). *ammistability*: Additive Main Effects and Multiplicative Interaction Model Stability Parameters. R package version 0.1.4.9000,  
<https://ajaygpb.github.io/ammistability/>, <https://CRAN.R-project.org/package=ammistability>.

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## Session Info

```
sessionInfo()
```

```
R Under development (unstable) (2025-12-09 r89129 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)
```

```
Matrix products: default
  LAPACK version 3.12.1
```

```
locale:
```

```
[1] LC_COLLATE=English_India.utf8  LC_CTYPE=English_India.utf8    LC_MONETARY=English_India.utf8
[4] LC_NUMERIC=C                    LC_TIME=English_India.utf8
```

```
time zone: Asia/Calcutta
```

```
tzcode source: internal
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

```
other attached packages:
```

```
[1] agricolae_1.3-7      ammistability_0.1.4.9000
```

```
loaded via a namespace (and not attached):
```

```
[1] generics_0.1.4      bitops_1.0-9        stringi_1.8.7       lattice_0.22-7      digest_0.6.39
[6] magrittr_2.0.4      evaluate_1.0.5      grid_4.6.0          RColorBrewer_1.1-3 fastmap_1.2.0
[11] plyr_1.8.9          tinytex_0.58        httr_1.4.7          pander_0.6.6        scales_1.4.0
[16] XML_3.99-0.20       Rdpack_2.6.4        cli_3.6.5           crayon_1.5.3        rlang_1.1.6
[21] rbibutils_2.4       withr_3.0.2         yaml_2.3.12         tools_4.6.0         parallel_4.6.0
[26] reshape2_1.4.5     AlgDesign_1.2.1.2   dplyr_1.1.4         ggplot2_4.0.1       mathjaxr_2.0-0
[31] curl_7.0.0         vctrs_0.6.5         R6_2.6.1            lifecycle_1.0.4     stringr_1.6.0
[36] MASS_7.3-65        cluster_2.1.8.1     pkgconfig_2.0.3     rJava_1.0-11        pillar_1.11.1
[41] ggcorrplot_0.1.4.1 gtable_0.3.6        rsconnect_1.7.0     glue_1.8.0          Rcpp_1.1.0.8.1
[46] xfun_0.54          tibble_3.3.0        tidyselect_1.2.1    rstudioapi_0.17.1   knitr_1.50
[51] dichromat_2.0-0.1  farver_2.1.2        htmltools_0.5.9     nlme_3.1-168        labeling_0.4.3
[56] rmarkdown_2.30     compiler_4.6.0      S7_0.2.1            RCurl_1.98-1.17
```

## References

- Ajay, B. C., Aravind, J., Abdul Fiyaz, R., Bera, S. K., Kumar, N., Gangadhar, K., et al. (2018). Modified AMMI Stability Index (MASI) for stability analysis. *ICAR-DGR Newsletter* 18, 4–5.
- Ajay, B. C., Aravind, J., and Fiyaz, R. A. (2019a). ammistability: R package for ranking genotypes based on stability parameters derived from AMMI model. *Indian Journal of Genetics and Plant Breeding (The)* 79, 460–466. doi:10.31742/IJGPB.79.2.10.
- Ajay, B. C., Aravind, J., Fiyaz, R. A., Kumar, N., Lal, C., Gangadhar, K., et al. (2019b). Rectification of modified AMMI stability value (MASV). *Indian Journal of Genetics and Plant Breeding (The)* 79, 726–731. Available at: <https://www.isgpb.org/article/rectification-of-modified-ammi-stability-value-masv>.
- Annicchiarico, P. (1997). Joint regression vs AMMI analysis of genotype-environment interactions for cereals in Italy. *Euphytica* 94, 53–62. doi:10.1023/A:1002954824178.
- Bajpai, P. K., and Prabhakaran, V. T. (2000). A new procedure of simultaneous selection for high yielding and stable crop genotypes. *Indian Journal of Genetics & Plant Breeding* 60, 141–146.
- Farshadfar, E. (2008). Incorporation of AMMI stability value and grain yield in a single non-parametric index (GSI) in bread wheat. *Pakistan Journal of biological sciences* 11, 1791. doi:10.3923/pjbs.2008.1791.1796.
- Farshadfar, E., Mahmodi, N., and Yaghotipoor, A. (2011). AMMI stability value and simultaneous estimation of yield and yield stability in bread wheat (*Triticum aestivum* L.). *Australian Journal of Crop Science* 5,



1837–1844.

- Gauch, H. G. (1988). Model selection and validation for yield trials with interaction. *Biometrics* 44, 705–715. doi:10.2307/2531585.
- Gauch, H. G. (1992). *Statistical Analysis of Regional Yield Trials: AMMI Analysis of Factorial Designs*. Amsterdam ; New York: Elsevier.
- Jambhulkar, N. N., Bose, L. K., Pande, K., and Singh, O. N. (2015). Genotype by environment interaction and stability analysis in rice genotypes. *Ecology, Environment and Conservation* 21, 1427–1430. Available at: [http://www.envirobiotechjournals.com/article\\_abstract.php?aid=6346&iid=200&jid=3](http://www.envirobiotechjournals.com/article_abstract.php?aid=6346&iid=200&jid=3).
- Jambhulkar, N. N., Bose, L. K., and Singh, O. N. (2014). “AMMI stability index for stability analysis,” in *CRRI Newsletter, January-March 2014*, ed. T. Mohapatra (Cuttack, Orissa: Central Rice Research Institute), 15. Available at: [https://crri.icar.gov.in/crnl\\_jan\\_mar\\_14\\_web.pdf](https://crri.icar.gov.in/crnl_jan_mar_14_web.pdf).
- Jambhulkar, N. N., Rath, N. C., Bose, L. K., Subudhi, H., Biswajit, M., Lipi, D., et al. (2017). Stability analysis for grain yield in rice in demonstrations conducted during rabi season in India. *Oryza* 54, 236–240. doi:10.5958/2249-5266.2017.00030.3.
- Purchase, J. L. (1997). Parametric analysis to describe genotype  $\times$  environment interaction and yield stability in winter wheat. Available at: <https://scholar.ufs.ac.za:8080/xmlui/handle/11660/1966>.
- Purchase, J. L., Hatting, H., and Deventer, C. S. van (1999). “The use of the AMMI model and AMMI stability value to describe genotype  $\times$  environment interaction and yield stability in winter wheat (*Triticum aestivum* L.),” in *Proceedings of the Tenth Regional Wheat Workshop for Eastern, Central and Southern Africa, 14-18 September 1998* (South Africa: University of Stellenbosch).
- Purchase, J. L., Hatting, H., and Deventer, C. S. van (2000). Genotype  $\times$  environment interaction of winter wheat (*Triticum aestivum* L.) in South Africa: II. Stability analysis of yield performance. *South African Journal of Plant and Soil* 17, 101–107. doi:10.1080/02571862.2000.10634878.
- Raju, B. M. K. (2002). A study on AMMI model and its biplots. *Journal of the Indian Society of Agricultural Statistics* 55, 297–322.
- Rao, A. R., and Prabhakaran, V. T. (2005). Use of AMMI in simultaneous selection of genotypes for yield and stability. *Journal of the Indian Society of Agricultural Statistics* 59, 76–82.
- Sneller, C. H., Kilgore-Norquest, L., and Dombek, D. (1997). Repeatability of yield stability statistics in soybean. *Crop Science* 37, 383–390. doi:10.2135/cropsci1997.0011183X003700020013x.
- Wricke, G. (1962). On a method of understanding the biological diversity in field research. *Zeitschrift für Pflanzenzüchtung* 47, 92–146.
- Zali, H., Farshadfar, E., Sabaghpour, S. H., and Karimizadeh, R. (2012). Evaluation of genotype  $\times$  environment interaction in chickpea using measures of stability from AMMI model. *Annals of Biological Research* 3, 3126–3136.
- Zhang, Z., Lu, C., and Xiang, Z. (1998). Analysis of variety stability based on AMMI model. *Acta Agronomica Sinica* 24, 304–309. Available at: <https://zwxb.chinacrops.org/EN/Y1998/V24/I03/304>.
- Zobel, R. W. (1994). “Stress resistance and root systems,” in *Proceedings of the Workshop on Adaptation of Plants to Soil Stress. 1-4 August, 1993. INTSORMIL Publication 94-2* (Institute of Agriculture; Natural Resources, University of Nebraska-Lincoln), 80–99.